

UNIVERSITY OF THE PHILIPPINES

Master of Science in Materials Science and Engineering

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Investigation of a-Si/SiO_2 antireflective coatings for terahertz pump wavelengths

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CHAPTER 1

INTRODUCTION

The development of devices which operate in the terahertz range has been a recent area of interest for researchers. The applications of terahertz waves include analysis of biological specimens [1], pharmaceuticals [2], and imaging [3] among others. The mechanisms by which these devices operate may include upconversion from the radio and mmWave range, or downconversion of photonic devices from the infrared region [4]. An example of a device which utilizes downconversion is the terahertz photoconductive antenna (PCA), and there are already multiple demonstrations and even commercial PCA devices that utilize the 780 nm and 1550 nm wavelengths, which are supplied by Ti:sapphire and telecommunication fiber lasers respectively [5], [6].

A PCA consists of a semiconductor material where the terahertz generation or absorption occurs and two metal electrodes on the semiconductor surface separated by a few microns that forms the photoconductive gap [7]. When a femtosecond pulse from the laser illuminates the gap, carriers are generated in the semiconductor and swept by an electric field from the bias applied to the metal electrodes, and the movement of the carriers generates the terahertz wave.

One of the common semiconductor materials of choice for a PCA is low temperature (LT) grown indium gallium arsenide (InGaAs), which has a bandgap of $E_g < 0.8$ eV and can be excited by telecommunications wavelength femtosecond fiber lasers operating at 1550 nm [8]. Another common semiconductor material used for PCAs is LT gallium arsenide (GaAs), which has a bandgap of $E_g < 1.42$ eV and operates at a much shorter wavelength of 780 nm[9] supplied by femtosecond Ti:sapphire lasers. However, when a bare PCA is illuminated by a laser there are significant reflection losses due to the high refractive index of the material[10]. Adding an antireflective coating to the PCA can increase the efficiency of the device by reducing the amount of light reflected by

the film [11]. The challenge lies in the design of a suitable antireflective coating for the aforementioned wavelengths.

1.1 Significance of the study

The development of antireflective coatings which reduce reflection losses at commonly used excitation wavelengths for terahertz devices holds potential for increasing the efficiency of terahertz (THz) PCA. By enhancing the efficiency of THz generation THz time domain spectroscopy (TDS) systems are improved, which in turn have diverse applications in fields such as biology, imaging, and pharmaceuticals.

The Semiconductors group of the Condensed Matter Physics Laboratory in UP Diliman is able to grow LT-InGaAs and LT-InGaAs layers via molecular beam epitaxy (MBE), fabricate them into PCAs via photolithography, and study them using 1550 nm and 780 nm wavelength lasers for THz TDS. However, the PCAs being developed lack antireflective coatings optimized for their operating wavelengths. By developing a mulitlayer antireflective coating stack composed of a-Si and SiO_2, two materials commonly used for CMOS processing, this study aims to address these problems, improving the THz intensity generated by the PCA and its overall performance. The findings could pave the way for more efficient THz devices.

1.2 Objectives of the study

The main objective of this study is to design, fabricate, and characterize an antireflective coating based on layers of a-Si and SiO_2 thin films that work at the excitation wavelengths commonly used for THz applications. Specifically, this will require the following objectives:

- to deposit a-Si and SiO_2 thin films using electron beam evaporation,
- to determine the material and optical properties of the deposited films,
- to design a multilayer antireflective coating of a-Si and SiO_2 tailored for the 1550 nm and 780 nm wavelengths using the transfer matrix method,

- to fabricate the multilayer antireflective coatings on bare Si wafers, and
- to measure the reflectance of the fabricated a-Si/SiO_2 antireflective coatings and compare them with the simulated reflectance.

1.3 Scope and limitations of the study

This study covers the development of an amorphous silicon (a-Si) and SiO_2 antireflective coating for the 1550 nm and 780 nm wavelengths which are commonly used for the excitation of LT-InGaAs and LT-GaAsTHz PCA. We have not included the integration of the developed antireflective coating with LT-InGaAs and LT-GaAsTHz PCAs, which requires more procedures such as the growth of these materials on GaAs or InP wafers via molecular beam epitaxy, patterning of the antenna via UV photolithography, deposition of metal contacts via electron beam lithography, and investigation of the emission via THz time-domain spectroscopy. We will also limit the variables for the deposition method of electron beam evaporation to the thickness of the a-Si and SiO_2 films and will not inlcude investigation of other growth parameters such as substrate heating and base pressure. The substrates used in this study are limited to polished Si wafers and glass, which are readily available and much cheaper compared to epi-ready semi-insulating GaAs. Our reflection measurements are also limited by the wavelength range of the existing Si photodetector (500 nm to 1000 nm).

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Electromagnetic Waves in Dielectric Materials

The propagation of electromagnetic waves through dielectric materials can be described using the Fresnel equations, which can be derived from the boundary conditions that constrain the electric and magnetic field components arising as a direct result of Maxwell's equations [12], [13]. Suppose incident light approaches a boundary between two dielectric materials with refractive indices n_1 and n_2 as shown in Figure 2.1. The first case shows s-polarized light, where the electric field is perpendicular to the plane of incidence, while the second case shows p-polarized light, where the electric field is parallel to the plane of incidence. The incident and reflected angles are equal $\theta_i = \theta_r$, and the transmitted angle is given by Snell's law $n_1 \sin\theta_i = n_2 \sin\theta_t$. The boundary conditions require that the components of the electric and magnetic fields be continuous, thus for s-polarized light, the electric fields are given by $E_i + E_r = E_t$ or $1 + r = t$, where r and t are the reflected and transmitted amplitudes and the incident amplitude is normalized to 1. The y-component of the magnetic field is given by $H_i \cos\theta_i - H_r \cos\theta_r = H_t \cos\theta_t$, which can be rewritten as $n_1 \cos\theta_i(1 - r) = n_2 \cos\theta_t t$ using the relation $H = nE/c$. Solving for r gives

$$r_s = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t} \tag{2.1}$$

while solving for t gives

$$t_s = \frac{2n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t}. \tag{2.2}$$

For p-polarized light, the magnetic fields are given by $H_i + H_r = H_t$ or $n_1(1+r) = n_2 t$, thus the electric fields are given by $E_i \cos\theta_i - E_r \cos\theta_r = E_t \cos\theta_t$, which can be

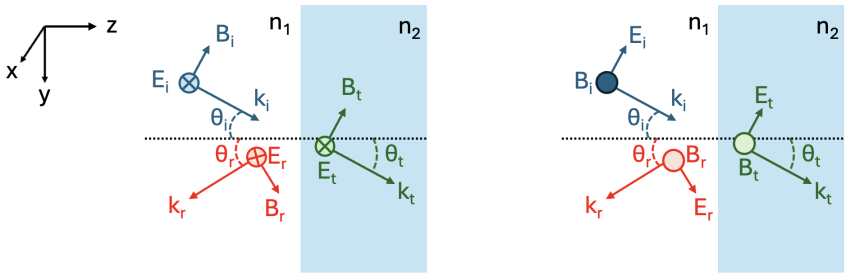


Figure 2.1: Diagram of polarized light at two interfaces. The left image shows the case for s-polarized light, with the electric field pointing into the page. For p-polarized light on the right the magnetic field points outside of the page.

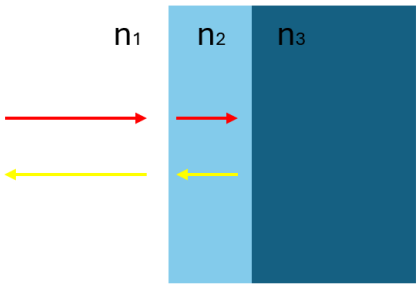


Figure 2.2: Diagram of light with normal incidence on a single film and substrate.

rewritten as $n_1(1-r)\cos\theta_i = n_2 t\cos\theta_t$. Solving for r gives

$$r_p = \frac{n_2\cos\theta_i - n_1\cos\theta_t}{n_2\cos\theta_i + n_1\cos\theta_t} \tag{2.3}$$

while solving for t gives

$$t_p = \frac{2n_1\cos\theta_i}{n_2\cos\theta_i + n_1\cos\theta_t}. \tag{2.4}$$

At normal incidence the reflectance R evaluates to

$$R = r_1 2^2 = \left|\frac{n_1 - n_2}{n_1 + n_2}\right|^2. \tag{2.5}$$

For an uncoated glass substrate ($n_{\text{glass}} \approx 1.5$) in air ($n_{\text{air}} \approx 1.0$), approximately 4% of incident light power is reflected per interface [14]. For high-index semiconductor materials like silicon or GaAs ($n_{\text{Si}} \approx 3.7$ at 780 nm), reflection losses exceed 30% [15].

Consider a single dielectric film of thickness d_2 and refractive index n_2 deposited between an ambient medium n_1 (air) and a substrate n_3, where $n_1 < n_2 < n_3$ as shown

in Figure 2.2. The reflectance of the structure at normal incidence would be given by

$$R = \frac{r_{12}^2 + r_{23}^2 + 2r_{12}r_{23}\cos(2\delta_2)}{1 + r_{12}^2 + r_{23}^2 + 2r_{12}r_{23}\cos(2\delta_2)} \tag{2.6}$$

where $\delta_2 = k_2 d_2 = 2\pi n_2 d_2/\lambda$ is the phase difference between the reflected light at the interfaces of the dielectric film. The $2r_{12}r_{23}\cos(2\delta_2)$ term oscillates, and at $\cos(2\delta_2) = \pm 1$ or when $2\delta_2$ is an integer multiple of 2π, interference of the reflected waves occur and the reflectance is at an extremum. The wavelength at the extremum is

$$\lambda = \frac{2n_2 d_2}{m} \tag{2.7}$$

where m is the number of cycles from one extremum to another. The refractive index of the thin film can be calculated if the thickness is known

$$n_2 = \frac{m\lambda_1\lambda_m}{2d_2(\lambda_m - \lambda_1)}. \tag{2.8}$$

and the new λ is the average of the two wavelengths $\lambda = (\lambda_m + \lambda_1)/2$.

The mechanism of classical thin-film antireflective coating (ARC)s relies on destructive wave interference. Two necessary conditions must be satisfied simultaneously to achieve zero reflectance ($R = 0$) at a specific target wavelength λ_t [16]. The first condition is that the optical path difference between the partial wave reflected from the top interface (r_1) and the partial wave reflected from the substrate interface (r_2) must result in a π phase shift ($180°$ out of phase). The physical film thickness d must satisfy:

$$d = \frac{\lambda_t}{4n_2}. \tag{2.9}$$

Since both reflection events occur at interfaces transitioning from a lower to a higher refractive index ($n_1 < n_2$ and $n_2 < n_3$), both undergo an phase change upon reflection. The round-trip optical path within the coating $2n_2 d_2 = \lambda_t/2$ introduces a net phase change of π, ensuring destructive interference.

The second condition for zero reflectance is that the intensities of the two reflected

waves must be identical $|r_1| = |r_2|$. Equalizing the Fresnel reflection coefficients yields:

$$\frac{n_2 - n_1}{n_2 + n_1} = \frac{n_3 - n_2}{n_3 + n_2} \implies n_2 = \sqrt{n_1 n_3}. \tag{2.10}$$

For a glass substrate ($n_3 = 1.5$) in air ($n_1 = 1.0$), the ideal coating index is $n_2 \approx 1.22$ [14]. However, finding natural bulk dielectric materials with $n \approx 1.22$ is difficult; magnesium fluoride (MgF_2, $n \approx 1.38$) is usually employed as an alternative [17]. Single-layer ARCs suffer from multiple limitations [18]. First, destructive interference is exact only at the design wavelength λ_t, with the reflection increasing rapidly as the wavelength gets further away. Another limitation of single-layer ARCs is that off-normal angles of incidence alter both the internal propagation path length and the polarization-dependent reflection coefficients, shifting the minimum reflection away from λ_t.

To achieve broadband and wide-angle antireflection, multi-layer dielectric stacks comprising alternating high-index and low-index thin films are used. Multi-layer structures introduce multiple internal reflection pathways that result in broader and more easily tailored destructive interference minima across the relevant spectrum. The design of multi-layer ARCs is usually performed computationally with the transfer matrix method[19].

2.2 Transfer Matrix Method

At the boundary between two dielectric materials, the electric field amplitudes of the incident, reflected, and transmitted waves can be related using a transfer matrix [20], [21]. In the schematic shown in Figure 2.3a, a wave with amplitude E_1^+ is moving to the right, and another wave with amplitude E_1^- is moving to the left. For the other side of the boundary, the amplitudes are E_2^+ and E_2^-. The fields can be characterized using two 2×1 matrices, one for each side of the boundary, and a 2×2 interface matrix $\mathbf{D}$ as shown:

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} D_{12} & D_{21} \\ D_{12} & D_{21} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \tag{2.11}$$

The values of the matrix elements D_{ij} can be obtained from inspecting Figure 2.3a. The subscripts i and j refer to the direction of the wave propagation, with t_{12} signifying transmission from medium 1 to medium 2, and r_{12} signifying reflection from medium 1 to medium 2. The fields are thus given by

$$E_2^+ = t_{12}E_1^+ + r_{21}E_2^- \tag{2.12}$$

$$E_1^- = r_{12}E_1^+ + t_{21}E_2^- \tag{2.13}$$

which can be rearranged to

$$E_1^+ = \frac{E_2^+ - r_{21}E_2^-}{t_{12}} \tag{2.14}$$

$$E_1^- = \frac{r_{12}}{t_{12}}\left(E_2^+ - r_{21}E_2^-\right) + t_{21}E_2^-. \tag{2.15}$$

Considering that the reflection coefficient is $r = (n_1 - n_2)\,/\,(n_1 + n_2)$ then $r_{12} = -r_{21}$. For the transmission coefficient $t = 2n_1/\,(n_1 + n_2)$, and $t_{12} = n_2 t_{21}/n_1$. Substituting these relations into Equation 2.14 and Equation 2.15 with the assumption that $R + T = 1$ yields the matrix equation for the interface $\mathbf{D}$

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \frac{1}{t_{12}} \begin{bmatrix} 1 & r_{12} \\ r_{12} & 1 \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix}. \tag{2.16}$$

The propagation matrix $\mathbf{P}$ for a layer of thickness d and refractive index n can be obtained by considering the geometry of the waves in Figure 2.3b, where a layer with refractive index n and thickness d has been inserted between the two waves. The wave moving to the right varies as $E^+ = E_0 e^{-j(k_y y + k_z z - \omega t)}$, and the difference between the two waves at the left and right boundaries is given by $E_2^+ = E_1^+ e^{jkz\cos\theta}$ with the reverse given by $E_2^- = E_1^- e^{-jkz\cos\theta}$. The propagation matrix is thus given by

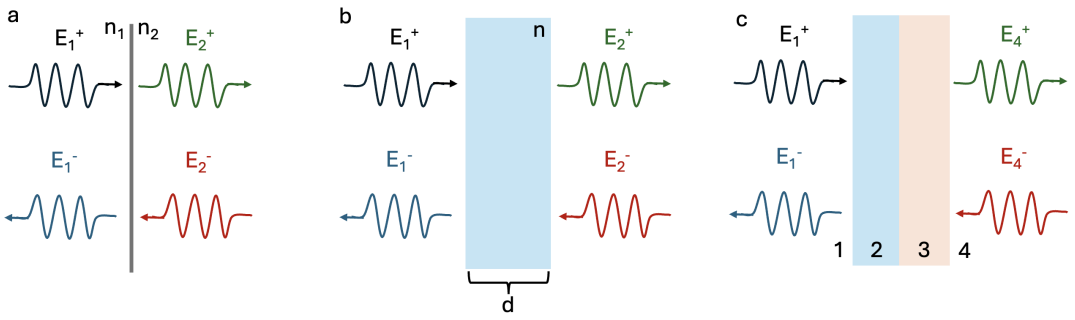


Figure 2.3: Propagation of waves at an interface (a), through a layer with refractive index n and thickness d (b), and through a multilayer system (c).

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} e^{jkd\cos\theta} & 0 \\ 0 & e^{-jkd\cos\theta} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \tag{2.17}$$

where $k = 2\pi n/\lambda$ is the wavenumber in the medium. This analysis can be extended to a two layer system by multiplying the interface and propagation matrices while considering the order of the layers. Starting from the rightmost boundary as shown in Figure 2.3c, the fields on both sides are related using the interface matrix by $\mathbf{E}_{3R} = \mathbf{D}_{34}\mathbf{E}_4$, where $3R$ is the field at the right side of layer 3. The fields at the left side of layer 3 are related to the right side by the propagation matrix $\mathbf{E}_{3L} = \mathbf{P}_3\mathbf{E}_{3R}$, which in turn can be related using another interface matrix to the field across the relevant boundary. This results in the following matrix equation relating the fields at the leftmost region to the rightmost region:

$$\mathbf{M} = \mathbf{D}_{12}\mathbf{P}_2\mathbf{D}_{23}\mathbf{P}_3\mathbf{D}_{34} \tag{2.18}$$

such that

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \mathbf{M} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix}. \tag{2.19}$$

This method allows the analysis of any multilayer system using matrix multiplication of the component interface $\mathbf{D}$ and propagation $\mathbf{P}$ matrices. The reflection coefficient r for a multilayer structure is simply

$$r = \frac{M_{21}}{M_{11}} \tag{2.20}$$

since $t = 1/M_{11}$ and $r = M_{21}/M_{11}$.

2.3 Electron beam physical vapor deposition of a-Si and SiO_2

Silicon and its native oxide SiO_2 are the most common materials used in the semiconductor industry. Thin films of Si and SiO_2 are used in microelectronics, photonics, and photovoltaic applications [22]. These thin films are grown using methods ranging from physical vapor deposition, chemical vapor deposition, to thermal oxidation. Electron beam physical vapor deposition (EBPVD) or e-beam evaporation is one commonly utilized technique for depositing high-purity Si and SiO_2 films due to its high deposition rate, precise thickness control, and capability to vaporize high-melting-point materials [23].

SiO_2 is widely utilized due to its wide bandgap of 8.9 eV, low refractive index of $n \approx 1.45 - 1.46$ in the visible spectrum, high chemical stability, and low dielectric loss for optical coatings [24]. While thermal oxidation can produce high-quality native SiO_2 from Si, it requires temperatures $>$900 °C and is restricted to silicon substrates [25]. Electron beam evaporation enables low-temperature deposition $<$300 °C on different substrates including polymers, glass, and compound semiconductors such as GaAs [26]. However, e-beam evaporation of SiO_2 presents unique physical challenges including dissociation during evaporation, non-stoichiometric SiO_2, porosity, and intrinsic film stress [27].

In e-beam evaporation, a high-energy electron beam is magnetically steered onto a high-purity target contained in a water-cooled crucible [23]. The electron generation is caused by Richardson-Dushman thermionic emission from a heated filament which is typically tungsten:

$$J = AT^2 \exp\left(-\frac{\Phi}{k_B T}\right) \tag{2.21}$$

where J is the emission current density A/m^2, $A \approx 1.2 \times 10^6$ A/(m^2 K^2) is the Richardson constant, Φ is the work function of the filament material in eV, k_B is the Boltzmann constant, and T is the absolute temperature in K. Emitted electrons are accelerated toward an anode using a high potential difference and the beam trajectory is controlled via the Lorentz force:

$$\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{2.22}$$

In a uniform transverse magnetic field $\mathbf{B}$, electrons follow a circular path. This bending caused by the magnetic field allows the electron gun to remain shielded from direct line-of-sight deposition from the source while enabling frequent sweeping across the crucible to maintain uniform evaporation from the source as shown in Figure 2.1. As the electron beam strikes the source, local temperatures at the beam spot may exceed $1700°$C, causing melting and evaporation for Si or direct sublimation in the case of SiO_2 [28]. Unlike pure Si, SiO_2 does not evaporate purely as intact SiO_2 molecules. Thermal dissociation may occur at the surface of the source, yielding a mixture of gaseous species:

$$\mathrm{SiO_2\,(s)} \longrightarrow \mathrm{SiO\,(g)} + \frac{1}{2}\mathrm{O_2\,(g)} \tag{2.23}$$

$$\mathrm{SiO_2\,(s)} \longrightarrow \mathrm{Si\,(g)} + \mathrm{O_2\,(g)} \tag{2.24}$$

Non-reactive e-beam evaporation of SiO_2 in high vacuum typically yields films of SiO_x with lower stoichiometric values $x < 2.0$. A lower stoichiometry impacts film performance by introducing sub-oxide states (Si^{1+}, Si^{2+}, Si^{3+}) and oxygen vacancies, leading to an increased optical absorption and increased refractive index. To ensure fully stoichiometric SiO_2, reactive e-beam evaporation can be performed by introducing oxygen gas into the chamber.

According to the structure zone model, the growth of the films proceed in three general steps [29]. The first step occurs when the vaporized molecules are transported towards the substrate, the second when they are adsorbed onto the surface of the substrate or growing film, their diffusion across the surface, and any chemical reactions that

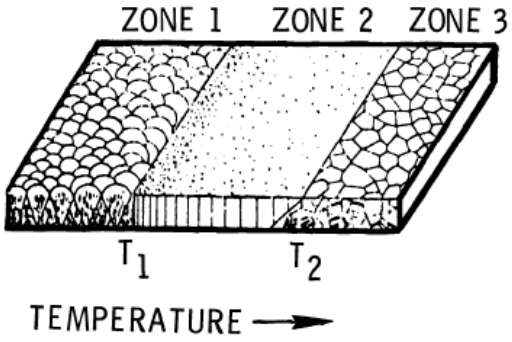


Figure 2.4: Structure zone diagram introduced by Movchan and Demchishin from study of vacuum evaporated metals and oxides.

may involve their incorporation into the film or removal via thermal desorption or sputtering. The third step involves the movement of the molecules to their final locations via bulk diffusion and solid state reactions.

Movchan and Demchishin classified film growth into distinct zones, with the morphology of the films depending on the normalized homologous temperature $T_h = T_{\text{sub}}/T_m$ where T_{sub} is the substrate temperature and T_m is the melting point of the material in K as shown in Figure 2.4 [30]. At low temperatures ($T_h < 0.3$) surface diffusion is low, and molecules essentially stick where they arrive, producing porous, tapered columnar structures. This is considered as Zone 1. At Zone 2 ($T_h < 0.5$) the temperatures are high enough to allow the surface diffusion fluxes to dominate over the arrival fluxes, but still low enough such that the bulk diffusion rates remain orders of magnitude less than the surface diffusion rates. When the temperature is high enough, bulk diffusion will dominate over all other processes and the formation is characterized by fully dense grains. This is considered as Zone 3. The morphology of the films also has an effect on the refractive index, with porous films generally having a lower index compared to the bulk.

2.4 Envelope method for refractive index

Accurate characterization of the optical properties of weakly absorbing thin films deposited on transparent substrates is crucial for integrated optical devices [31]. Building

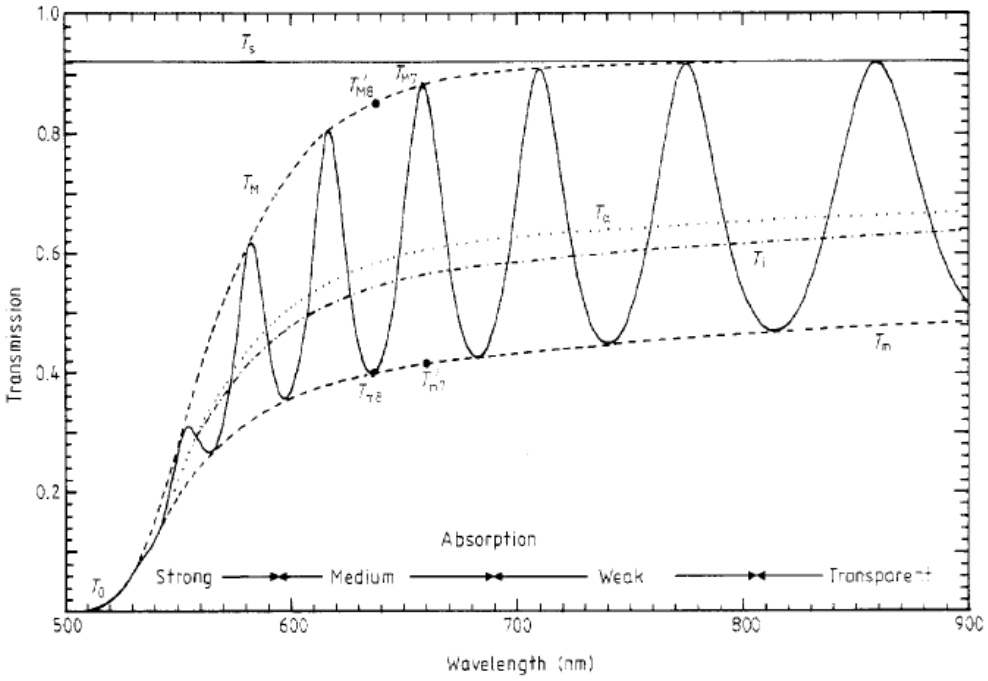


Figure 2.5: Envelope method as described by Swanepoel for a 1 µm film of a-Si:H on a finite glass substrate with transmittance T_s.

upon earlier developed interference envelope concepts [32], Swanepoel formulated a method which relies solely on the upper (T_M) and lower (T_m) continuous envelope curves fitted to the fringe extrema of a single normal-incidence transmission spectrum $T(\lambda)$ as shown in Figure 2.5 [33]. Consider a homogeneous thin dielectric film of uniform physical thickness d, complex refractive index $\tilde{n} = n - ik$, deposited on a transparent, thick substrate of index s surrounded by air ($n_0 = 1$) as shown in Figure 2.2. In the weakly absorbing region ($k^2 \ll n^2$), internal multiple reflections give rise to alternating constructive and destructive interference fringes in the transmission spectrum $T(\lambda)$.

The optical transmittance T at normal incidence, accounting for incoherent reflections at the rear substrate boundary, is given by:

$$T = \frac{Ax}{B - Cx\cos(\phi) + Dx^2} \tag{2.25}$$

where:$x = \exp(-\alpha d)$ is the absorptance term, and $\alpha = 4\pi k/\lambda$ is the absorption coefficient, $\phi = 4\pi nd/\lambda$ is the phase shift per round trip in the film. The system constants

A, B, C, and D depend strictly on n and s:

$$A = 16n^2 s$$

$$B = (n+1)^3(n+s^2)$$

$$C = 2(n^2-1)(n^2-s^2)$$

$$D = (n-1)^3(n-s^2)$$

where $s = 1/T_s + \sqrt{1/T_s - 1}$ is the refractive index of the substrate and T_s the transmittance of the substrate. The transmission spectrum exhibits local maxima T_M when at conditions of constructive interference $\cos(\phi) = +1$ and local minima T_m when $\cos(\phi) = -1$ or destructive interference. Continuous envelope curves $T_M(\lambda)$ and $T_m(\lambda)$ across the spectrum can be created using various functions, such as the McClain algorithm [34], parabolic or exponential interpolation [35].

For exponential interpolation, the envelope follows the following mathematical expression:

$$f^{\exp}(\lambda, T_0, T_1, T_2, \Lambda_1, \Lambda_2) = T_0 + T_1 e^{-\lambda/\Lambda_1} + T_2 e^{-\lambda/\Lambda_2} \tag{2.26}$$

where the optimal parameters $(\lambda, T_0, T_1, T_2, \Lambda_1, \Lambda_2)$ are obtained by solving a nonlinear-least squares fitting problem. Although this method is less precise than other algorithms, it has the benefit of being more appropriate for transmission spectra with a limited number of oscillations and tangent points, which often occurs for ultra-thin films. Common interpolations typically require at least four tangent points for a reliable estimation of each envelope [34], [35].

By setting $\cos(\phi) = \pm 1$ in the fundamental transmittance equation, an explicit equation for the refractive index n at any continuous wavelength λ in the transparent/weakly absorbing regime ($x \approx 1$) can be derived [33]:

$$n = \left[N + \sqrt{N^2 - s^2}\right]^{1/2} \tag{2.27}$$

where

$$N = 2s\left(\frac{T_M - T_m}{T_M T_m}\right) + \frac{s^2+1}{2} \tag{2.28}$$

From the interference condition, the fundamental equation for fringe order m is:

$$2nd = m\lambda \tag{2.29}$$

where m is an integer for maxima (T_M) and a half-integer for minima (T_m). If n_a and n_b are the calculated refractive indices at two adjacent extrema wavelengths λ_a and λ_b then thickness can be determined:

$$d = \frac{\lambda_a \lambda_b}{2(\lambda_a n_b - \lambda_b n_a)} \tag{2.30}$$

Taking the average over multiple adjacent fringe pairs yields a more accurate value for physical thickness d [33], [36]. Once an initial value of d is obtained, exact integer and half-integer values of m_0 are determined by evaluating Equation 2.29. Fitting these rounded integers back into Equation 2.29 gives more refined values of n and d for the extrema. The resulting values of $n(\lambda)$ can be fitted to Cauchy's equation to obtain a continuous function of n with the wavelength:

$$n(\lambda) = a + \frac{b}{\lambda^2} \tag{2.31}$$

2.5 X-ray reflectivity

As multi-layer optical stacks require nanometer precision, non-destructive surface characterization techniques are essential [37], [38]. X-ray reflectometry (XRR) is a non-destructive optical characterization technique widely employed to determine the structural properties of thin films and multi-layer heterostructures with nanometer resolution. By measuring the speculatively reflected intensity of X-rays as a function of incident angle, XRR provides absolute determinations of film thickness d, electron density ρ_e, and interface or surface roughness σ. For hard X-rays $\lambda \approx 0.05 - 0.2$ nm, the optical

refractive index n of condensed matter is slightly less than unity:

$$n = 1 - \delta - i\beta \tag{2.32}$$

where δ represents the dispersion term, which governs phase velocity and wave deflection and β represents the absorption term, which governs wave attenuation.

In terms of microscopic atomic parameters, δ and β are directly linked to the bulk properties of the material [38]:

$$\delta = \frac{r_e \lambda^2}{2\pi} \rho_e = \frac{r_e \lambda^2 N_A}{2\pi} \sum_j \frac{\rho_m (f_j^0 + f_j')}{M_j} \tag{2.33}$$

$$\beta = \frac{\mu \lambda}{4\pi} = \frac{r_e \lambda^2 N_A}{2\pi} \sum_j \frac{\rho_m f_j''}{M_j} \tag{2.34}$$

where $r_e = e^2/4\pi\varepsilon_0 m_e c^2 \approx 2.818 \times 10^{-15}$ m is the classical electron radius, ρ_e is the electron density of the material, N_A is Avogadro's number, ρ_m is the mass density, and M_j is the atomic molar mass, $(f_j^0 + f_j' + i f_j'')$ represents the complex atomic scattering factor, and μ is the linear x-ray attenuation coefficient. Because δ is positive and typically on the order of 10^{-6} to 10^{-5}, the real part of n $(1 - \delta)$ is less than 1.0. Consequently, vacuum or air behaves as an optically denser medium than any solid or liquid in the X-ray regime.

Because $n < 1$, X-rays impinging on a material surface from air/vacuum ($n_0 \approx 1$) undergo total external reflection when the incident grazing angle θ is below a critical angle θ_c. Applying Snell's Law ($\cos\theta_c = n = 1 - \delta$):

$$\cos\theta_c = 1 - \delta \tag{2.35}$$

Using the Taylor expansion $\cos\theta_c \approx 1 - \theta_c^2/2$ for small angles:

$$\theta_c \approx \sqrt{2\delta} = \lambda \sqrt{\frac{r_e \rho_e}{\pi}} \tag{2.36}$$

At angles $\theta \leq \theta_c$, 100% of the incident X-ray intensity is reflected ($R = 1$), and the

electromagnetic wave inside the material decays exponentially as an evanescent wave with a penetration depth of only a few nanometers. Above θ_c, the reflectivity drops sharply, decaying as θ^{-4} according to Fresnel's law [37].

When an X-ray beam strikes a thin film deposited on a substrate of different electron density, waves reflect from both the top interface of air and film and the bottom interface of the film and substrate. Constructive and destructive interference between these coherent wave paths produces periodic oscillations in reflectivity known as Kiessig fringes [39]. The fringe spacing $\Delta\theta$ is inversely proportional to the film thickness d:

$$d \approx \frac{\lambda}{2\Delta\theta}. \tag{2.37}$$

Real surfaces exhibit physical roughness (σ) and interdiffusion across boundaries. Roughness scatters X-rays non-specularly, dampening the reflected amplitude.The reduction in specular reflectivity due to root-mean-square (RMS) roughness σ is modeled using the Nevot-Croce factor [40], which modifies the ideal Fresnel reflection coefficient $r_{j,j+1}$:

$$r_{j,j+1}^{\text{rough}} = r_{j,j+1}^{\text{ideal}} \exp\left(-2k_{z,j}k_{z,j+1}\sigma_{j,j+1}^2\right) \tag{2.38}$$

where $k_{z,j} = 2\pi n_j \sin\theta_j/\lambda$ is the z-component of the wave vector in layer j. Increased roughness causes a faster drop-off in total reflectivity and dampens Kiessig fringe amplitudes. The effect of the three parameters on the XRR intensity is shown in Figure 2.6.

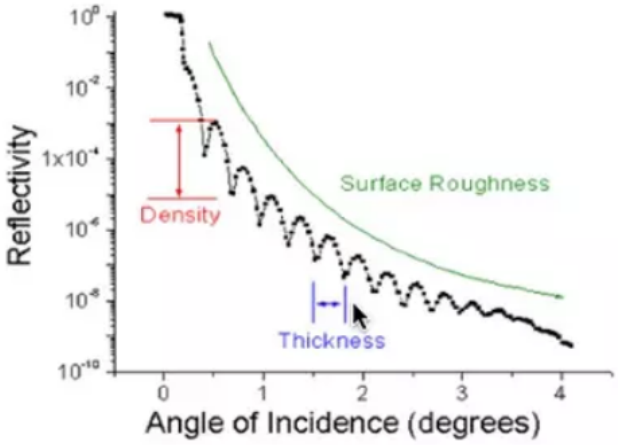


Figure 2.6: Effects of thickness, surface roughness and difference in density on the XRR intensity

2.6 Terahertz photoconductive antenna

The range of frequencies between microwave and infrared in the electromagnetic spectrum is known as the THz range. It has also been called the THz gap previously due to the difficulty of finding emitters and detectors which worked in this frequency range[41]. THz radiation is non-ionizing and can penetrate through organic material such as plastic and clothing, thus it has potential use in biological characterization[1], detection of contraband materials[42], and chemical identification[43]. Some substances have specific THz absorption spectra which can be used to identify their composition, as shown in Figure 2.7. The interaction amongst the bonds and forces in the molecules result in vibrational modes visible in the THz region[42]. On the other hand, THz reflection can be used to detect concealed objects when the concealing material is transparent to THz radiation, as the case for dielectrics such as clothing[44]. Another common applications for THz is time-domain spectroscopy or TDS.

In THz TDS, a THz pulse is generated using a PCA. The shape of the electrodes influences the pattern of the THz radiation generated by the PCA. Dipole and bowtie patterns demonstrate higher polarization ratios compared to spiral antennas while spiral and log-spiral antennas have a higher calculated radiation efficiency compared to bowtie and dipole antennas[45]. The shapes of the antennas are presented in Figure 2.8.

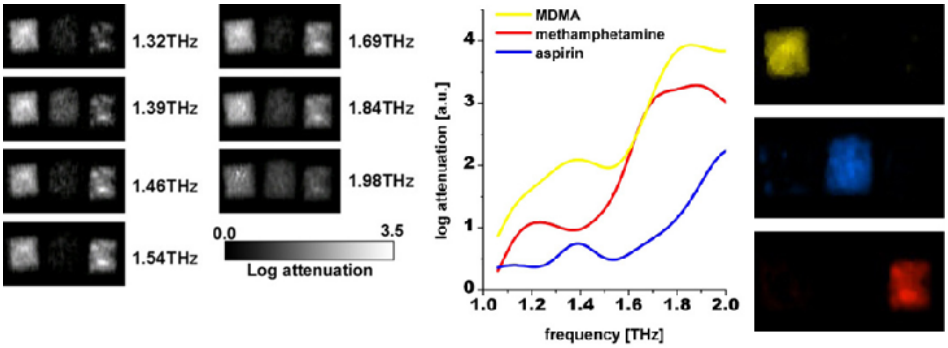


Figure 2.7: THz identification of MDMA, methamphetamine, and aspirin through a paper envelope. The substances are differentiated using their THz spectra[42]

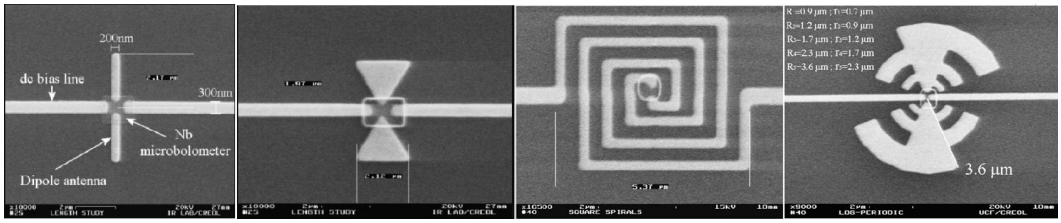


Figure 2.8: Commonly used THz PCA geometries; from left to right: dipole, bowtie, spiral, and log-spiral[45]

The emission mechanism for the PCA is as follows: the photoconductive material between the electrodes is illuminated with a femtosecond laser pulse, which generates photocarriers that are swept by an electric field applied to the electrodes of the PCA[7], [8]. The transient current $i_{PC}(t)$ resulting from the moving carriers generates electromagnetic radiation according to Maxwell's equations, with the intensity of the electric field component of the THz wave radiated by a dipole antenna expressed byEquation 2.39

$$E_{THz}(r,\theta,t) = \frac{l_e}{4\pi\varepsilon c^2 r}\frac{\partial i_{PC}(t)}{\partial t}\sin(\theta) \tag{2.39}$$

where l_e is the antenna length, ε the permittivity, c the speed of light, r the distance from the dipole, and θ the angle relative to the dipole axis[7].

The transient current density term $i_{PC}(t)$ is proportional to the convolution integral of the carrier density $n(t)$ and electron velocity $v_e(t)$ as shown in Equation 2.40[7].

$$i_{PC}(t) \propto P_{pulse}(t) \otimes (n(t)\, v_e(t)). \tag{2.40}$$

The carrier density is described by

$$\frac{dn}{dt} = \eta \frac{P_{\text{pulse}}(t)}{h\nu} - \frac{n}{\tau_c} \tag{2.41}$$

and using a 1D Drude-Lorentz model, the electron velocity by

$$\frac{dv_e}{dt} = -\frac{v_e}{\tau_s} + \frac{e}{m^*} E \tag{2.42}$$

where η is the absorption efficiency, $P_{\text{pulse}}(t)$ the ultrashort laser pulse power considered over time, τ_c, τ_c, and m^* the carrier lifetime, momentum relaxation time, electron effective mass respectively, and E the local electric field[7].

A photoconductive antenna is also used to detect THz radiation, with the incoming THz pulse focused together with a femtosecond pulse split from the same source at the gap of the antenna. As the femtosecond probe pulse arrives at the gap, carriers are excited and then accelerated proportionally by the electric field provided by the incoming THz pulse[8]. The difference of the probe pulse and the THz pulse provides a measurement in the time domain for the transient of the THz pulse. A schematic of a complete THz TDS system with emitter and detector is shown in Figure 2.9. Some of the photoconductive materials used for THz TDS systems include LT grown GaAs and InGaAs[46], [47], [48]. LT-GaAs is grown via molecular beam epitaxy at $200\,^{\circ}\text{C}$ to $300\,^{\circ}\text{C}$ instead of the usual growth temperature around $600\,^{\circ}\text{C}$[49].

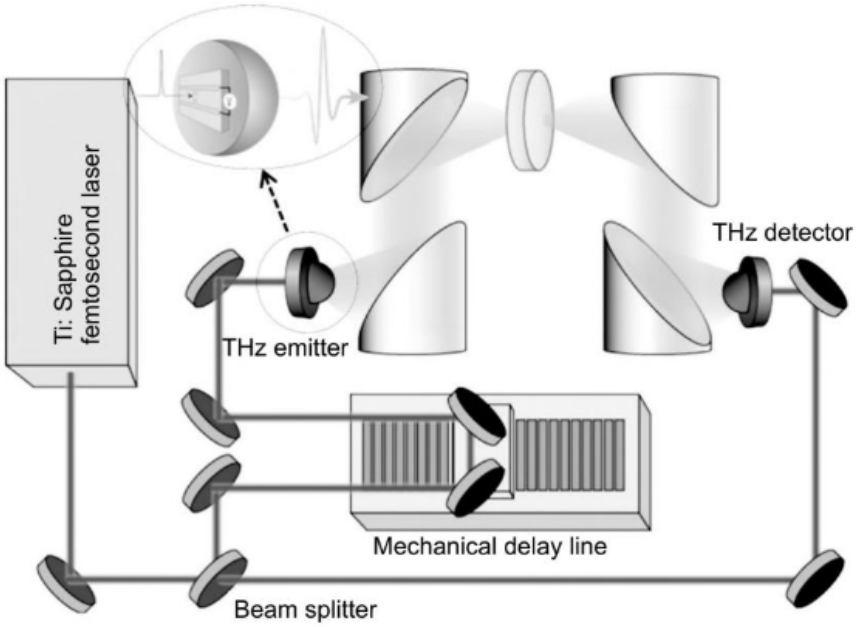


Figure 2.9: Schematic of a THz TDS setup with a femtosecond laser and two photoconductive antennas[7]

CHAPTER 3

METHODOLOGY

3.1 Initial growth of a-Si and SiO_2 thin films via electron beam evaporation

Glass slides and Si wafers were used as substrates for the initial growth via electron beam evaporation of a-Si and SiO_2 thin films respectively. The substrates were cleaned by sonication in acetone, isopropanol, and deionized water for 5 minutes each, after which they were blow-dried with nitrogen gas. The substrates were then loaded into the ULVAC electron beam evaporation chamber, where a base pressure of $4.5 \times 10^{-4}\,\mathrm{Pa}$ was achieved before actual deposition. Deposition was carried out at room temperature, and the nominal deposition rates and thicknesses of the films were monitored using a ULVAC CRTM-6000G deposition controller quartz crystal microbalance (QCM), with input values of z-factor $= 0.712$ and $\rho = 2.32\,\mathrm{g/cm^3}$ for Si and z-factor $= 1$ and $\rho = 2.65\,\mathrm{g/cm^3}$ for SiO_2. A reliable method to measure the thickness of the thin films in real time during deposition is necessary, which can be provided by the QCM.

3.2 Material and optical characterization of a-Si and SiO_2 thin films

After unloading the samples from the chamber, the XRR spectra of the films were obtained using a Malvern Panalytical Empyrean x-ray diffractometer. The thicknesses of the films were calculated using both the direct method and the Fourier method. The direct method involves manual counting of the fringes and dividing the calculated thickness from the angle difference of the selected extrema, whereas the Fourier method can be performed directly by the Advanced Material Analysis and Simulation Software which comes with the diffractometer. Curve fitting was also performed on selected samples to obtain the densities and interface roughness of the films.

The transmission spectra of the a-Si film on glass substrate was obtained via UV-Vis

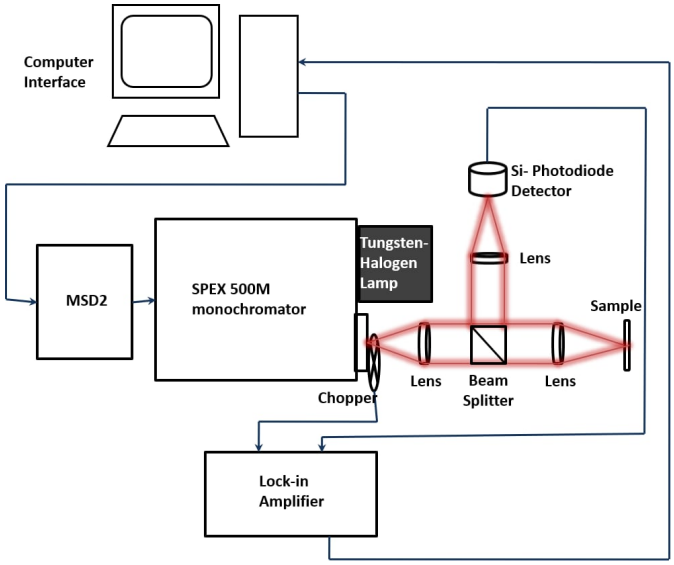


Figure 3.1: Normal incidence reflectance spectroscopy setup.

spectroscopy with a Shimadzu UV-Vis spectrophotometer while the reflectance spectra of the SiO_2 film on Si(100) wafer was measured via the reflectance spectroscopy setup as shown in Figure 3.1. Two methods were utilized to obtain the refractive indices as a function of wavelength for each film.

For a-Si, the first method used a modified version of the envelope method described in section 2.4 which yields $n(\lambda)$ at the extrema points that can be fitted to the Cauchy equation. The second method involved fitting the Cauchy equation directly to the transmission curve using the transfer matrix method to obtain $n(\lambda)$ over the whole range of wavelengths.

For SiO_2, the first method involved identification of the refractive indices from the reflectance spectra using the interference method described in section 2.1, which also yields $n(\lambda)$ at specified points that can be fitted to the Cauchy equation, and the second method involves fitting the Cauchy equation directly to the reflectance spectra using the transfer matrix method to obtain $n(\lambda)$ over the whole wavelength range.

3.3 Transfer matrix simulation of multilayer a-Si/SiO_2 antireflective coating

A transfer matrix method (TMM) implementation in Python was used to calculate the optimal thicknesses of the Si and SiO_2 layers for the antireflective coating stack. The

refractive indices of the a-Si and SiO_2 layers used for the simulation were obtained from the previous section. The optimal thicknesses of the layers were calculated to minimize the reflectance at $1550\,\mathrm{nm}$ and $780\,\mathrm{nm}$, which are the wavelengths of the lasers which can be used in the THz-TDS spectroscopy system. The program first iterated over a 100 nm range with 5 nm step size for each layer while calculating the reflectance of the stack, selecting the thicknesses that resulted in the lowest sum of the reflectances at the target wavelengths. A finer sweep with 10 nm range and 1 nm step size followed the rough sweep then the program outputs the optimized thickness. The number of layers and sequence of the stack was manually set by the user, prioritizing the least number of layers and the smallest thickness while still keeping the reflectivity less than 1 %.

3.4 Fabrication and characterization of a-Si/SiO_2 antireflective coating

The thicknesses obtained from the simulation were then used as parameters for the electron beam physical vapor deposition of the a-Si/SiO_2 antireflective coating on bare wafers. After deposition, the reflectance of the samples with the antireflective coating was measured via reflectance spectroscopy and compared to the simulated reflectance to verify the antireflective properties of the multilayer stack.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Optical and material properties of a-Si and SiO_2 thin films

4.1.1 a-Si thin films

Visual inspection

Figure 4.1 shows the a-Si thin films of 0.138 kÅ and 3.050 kÅ thickness as measured by the QCM on glass substrates together with a bare glass substrate. Compared to the bare glass, the substrates with a-Si films have a darker tint that intensifies as the thickness of the layer increases due to the absorption of Si in the visible range, confirming the deposition of a-Si.

Thickness, density and roughness

The QCM is able to give a nominal value of the film thickness during growth but it requires validation which can be provided by x-ray reflectivity (XRR) measurements. Figure 4.2 shows the XRR spectra of several a-Si thin films with varying thicknesses according to the QCM reading during the deposition process. Table 4.1 summarizes the QCM and XRR thicknesses.

The Kiessig fringes discussed in section 2.5 are apparent from Figure 4.2 and their

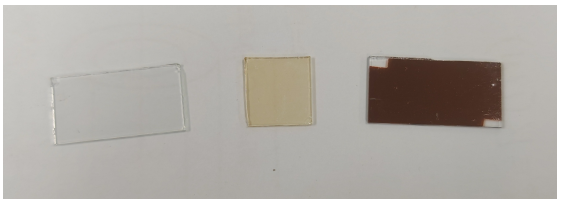
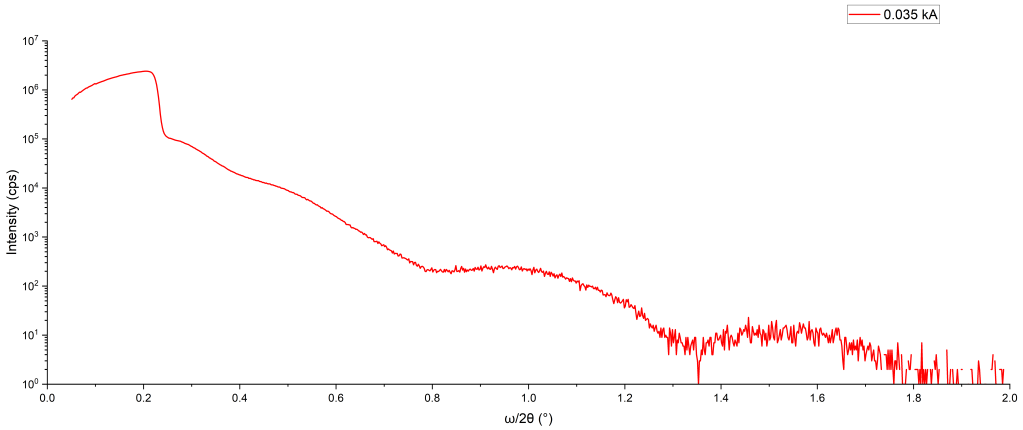
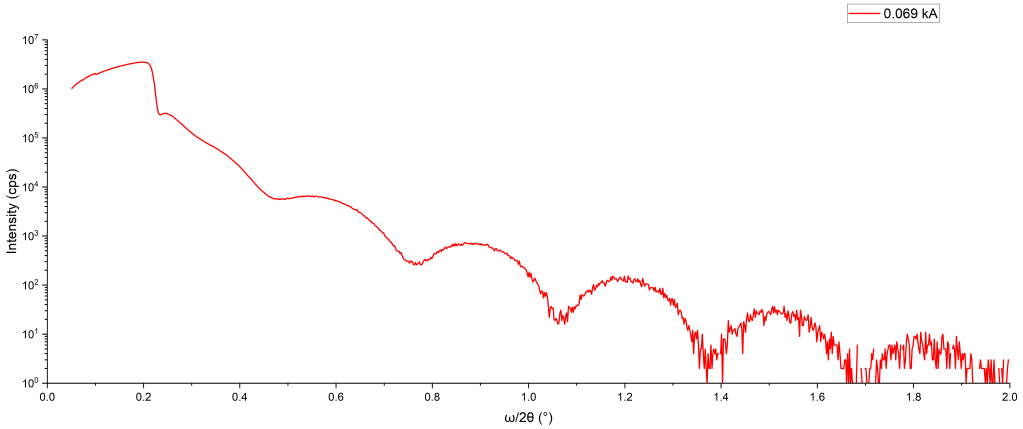


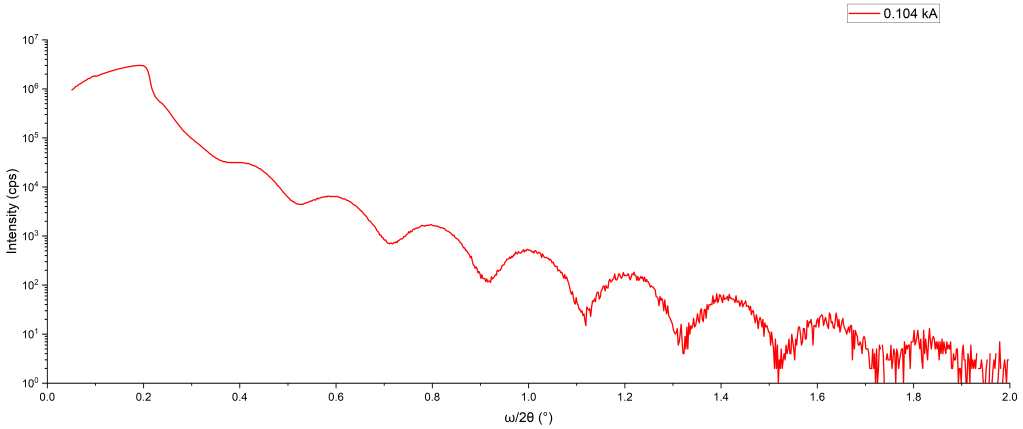
Figure 4.1: Bare glass, 0.138 kÅ and 3.050 kÅ thick a-Si films on glass. The intensity of the tint increases with the deposited a-Si thickness due to the absorption of Si in the visible range.



(a) 0.035 kÅ a-Si thin film



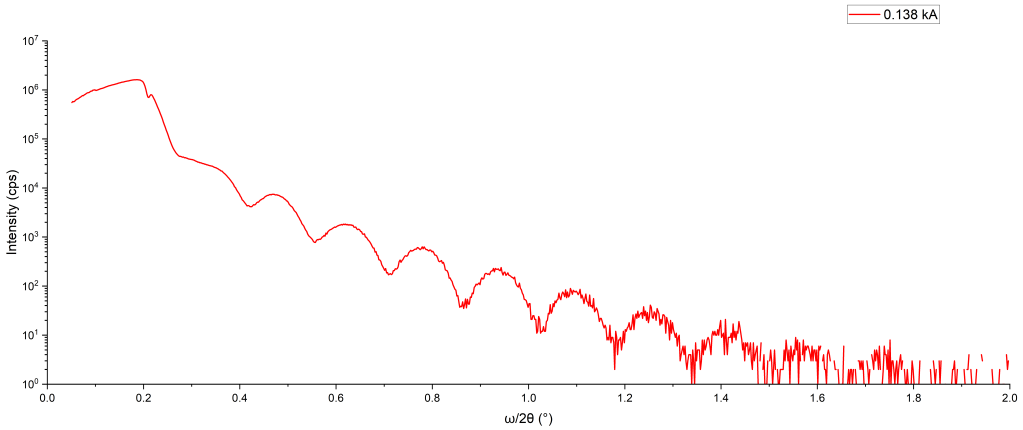
(b) 0.069 kÅ a-Si thin film



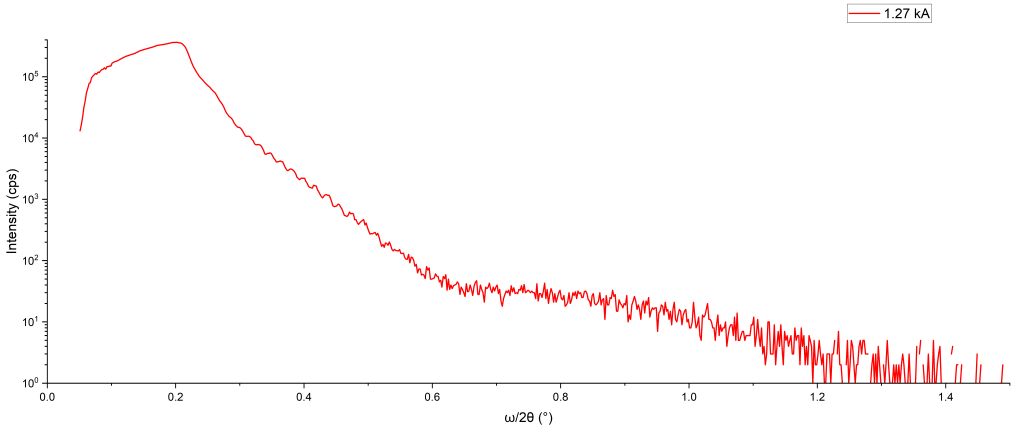
(c) 0.104 kÅ a-Si thin film

Figure 4.2: XRR measurements in log scale of a-Si thin films with varying QCM thicknessses. Kiessig fringes are visible, with the length of the fringes inversely proportional to the thickness of the film.

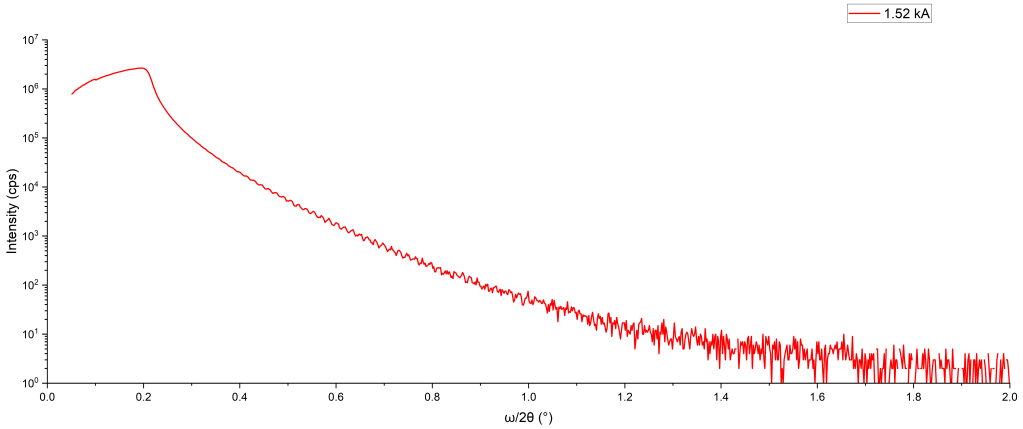
size varies depending on the QCM thickness. The largest fringes appear in the 0.035 kÅ thick film and get progressively smaller as the QCM thickness reading increases, with



(d) 0.138 kÅ a-Si thin film



(e) 1.270 kÅ a-Si thin film



(f) 1.520 kÅ a-Si thin film

Figure 4.2: XRR measurements of a-Si thin films, continued

the 1.520 kÅ film having very small fringes. No fringes could be identified for films beyond this thickness, which is attributed to the limited resolution of the x-ray diffractometer. The step size of the machine is , which implies that any fringes smaller than

QCM (kÅ)	Direct (nm)	Fourier (nm)
0.035	7.822	6.90
0.069	14.077	13.80
0.104	21.290	21.20
0.138	27.529	27.50
1.270	214.620	213.15
1.520	259.840	259.20

Table 4.1: a-Si thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra. The results of the two methods are generally in agreement with each other.

	Thickness (nm)	Roughness (nm)	Density (g/cm^3)
21.20 nm a-Si	21.9115	0.7047	1.80726
27.50 nm a-Si	28.4339	0.8008	1.75748

Table 4.2: Thickness, roughness and densities calculated from XRR curve fitting

this can no longer be resolved. In Table 4.1 the results of the direct and Fourier method are in general agreement, signifying the accuracy of the XRR method in determining the thickness of the film.

Linear regression of the XRR and QCM thicknesses as shown in Figure 4.3 gives

$$y = 168.03x + 3.0403 \tag{4.1}$$

with $R^2 = 0.9999$, where y is the measured XRR thickness in nm and x is the QCM reading in kÅ. The high R^2 value signifies that this equation relating the QCM thickness to the XRR thickness is reliable and can be used to match the real-time QCM readings during deposition to the actual thickness of the deposited film.

The results of the curve fitting performed using the Advanced Material Analysis and Simulation Software for the XRR spectra of selected a-Si thin film thicknesses as shown in Figure 4.4 and Figure 4.5 are presented in Table 4.2. The thicknesses match the previously calculated values in the direct and Fourier method. The calculated roughness for both films is less than 1 nm, signifying the excellent quality and sharp interface of the deposited film required for multilayer antireflective coatings. The densities of the films are generally lower than the expected density for silicon as used for the QCM

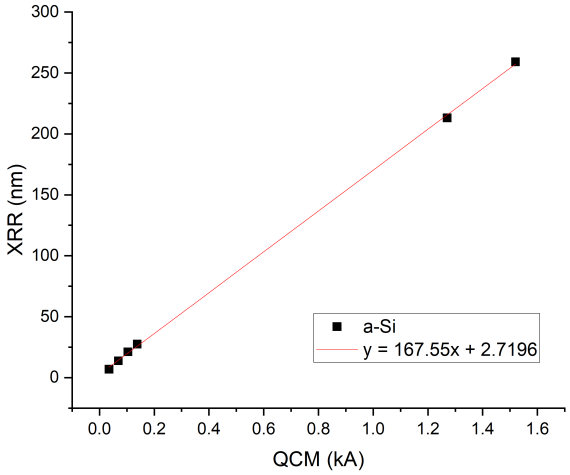


Figure 4.3: Linear regression of XRR and QCM thicknesses for a-Si thin films grown via e-beam evaporation. The high R^2 value indicates the validity of the equation in relating the QCM thickness to the actual thickness as measured by XRR.

input ($\rho = 2.32$). This difference is likely caused by the increased porosity of the films during deposition resulting from the absence of substrate heating. A higher substrate temperature promotes migration of the atoms resulting in increased density and crystallization as discussed in section 2.3.

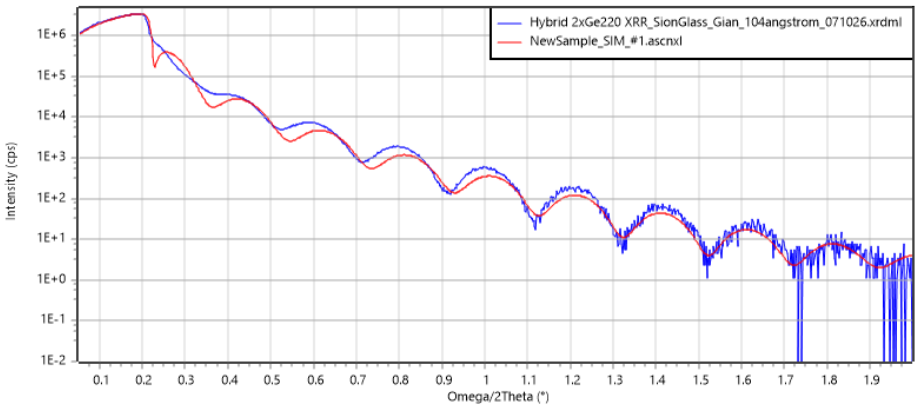


Figure 4.4: XRR curve fit to 21.20 nm a-Si thin film

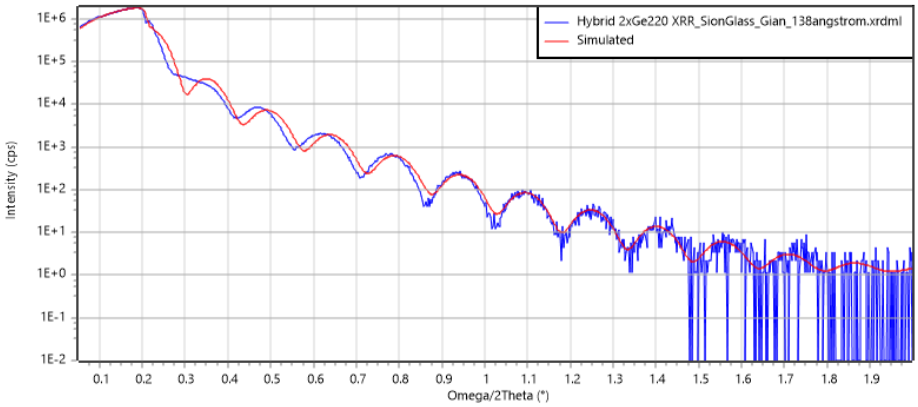


Figure 4.5: XRR curve fit to 27.50 nm a-Si thin film

4.1.2 SiO_2 thin films

Visual inspection

Figure 4.6 shows the SiO_2 thin films of 2.550 kÅ and 5.000 kÅ thickness as measured by the QCM on polished $Si(100)$ substrates together with a bare $Si(100)$ wafer. Compared to the bare wafer with a metallic color, the substrates with SiO_2 films have a color that changes depending on the thickness of the layer caused by thin film interference, confirming the deposition of SiO_2.

Thickness, density and roughness

Figure 4.7 shows the XRR spectra of several SiO_2 thin films with varying thicknesses according to the QCM reading during the deposition process. Table 4.3 summarizes the QCM and XRR thicknesses.

Similar to the previous section, the Kiessig fringes are visible in Figure 4.7 and their

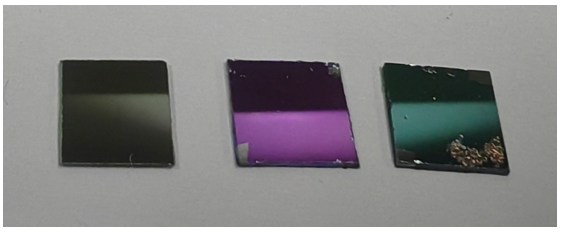
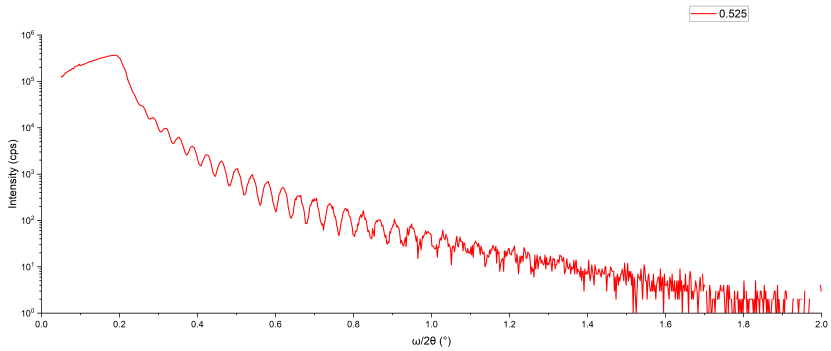
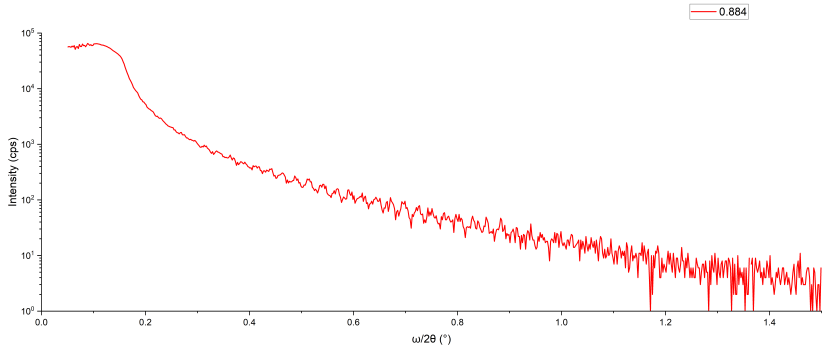


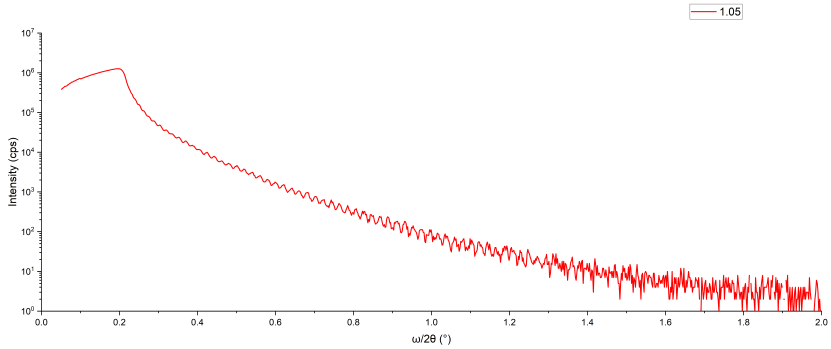
Figure 4.6: Bare Si wafer, 2.550 kÅ and 5.000 kÅ thick SiO_2 films on glass. The color of the wafer changes depending on the thickness of the deposited SiO_2.



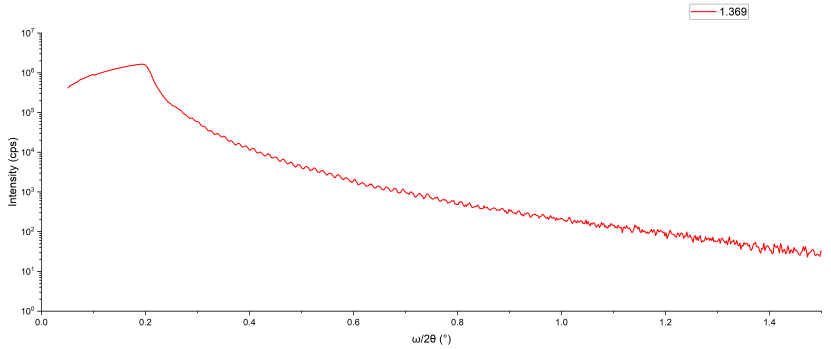
(a) 0.525 kÅ SiO_2 thin film



(b) 0.884 kÅ SiO_2 thin film



(c) 1.050 kÅ SiO_2 thin film



(d) 1.369 kÅ SiO_2 thin film

Figure 4.7: XRR measurements in log scale of SiO_2 thin films with varying QCM thicknessses. Kiessig fringes are visible, with the length of the fringes inversely proportional to the thickness of the film.

QCM (kÅ)	Direct (nm)	Fourier (nm)
0.525	104.6	103.9
0.884	165.1	163.2
1.050	206.6	204.7
1.369	272.3	270.1

Table 4.3: SiO_2 thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra

size varies depending on the QCM thickness. The largest fringes appear in the 0.525 kÅ thick film and get progressively smaller as the QCM thickness reading increases, with the 1.369 kÅ SiO_2 film having very small fringes. No fringes could be identified for films beyond this thickness, which we attribute to the limited resolution of the x-ray diffractometer. The step size of the machine is , which implies that any fringes smaller than this can no longer be resolved. In Table 4.3 the results of the direct and Fourier method are in general agreement, signifying the accuracy of the XRR method in determining the thickness of the film.

Linear regression of the XRR and QCM thicknesses as shown in Figure 4.8 gives

$$y = 198.52x - 4.5056 \tag{4.2}$$

with $R^2 = 0.9941$, where y is the measured XRR thickness in nm and x is the QCM reading in kÅ. Even though the value is less than that of the a-Si films in the previous section, the high R^2 value signifies that this equation relating the QCM thickness to the XRR thickness is also reliable and can be used to match the real-time QCM readings during deposition to the actual thickness of the deposited film. The lower R^2 value in the regression for SiO_2 might be caused by muliple factors, some of which could be increased porosity, a difference in stoichiometry of the film versus the source (i.e SiO_x where $x < 2$), or the variation in the deposition rate during the growth of the SiO_2 films, which in turn may be caused by the difficulty in maintaining a stable deposition rate for SiO_2 due to its low thermal conductivity and tendency to sublime instead of melting into a uniform liquid pool during heating by the electron beam as is the case for other

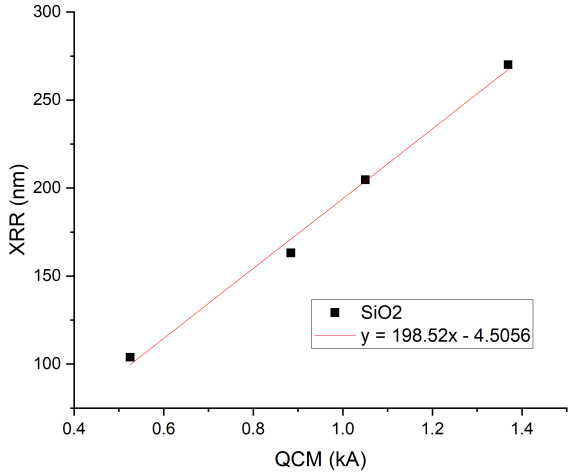


Figure 4.8: Linear regression of XRR and QCM thicknesses for SiO_2 thin films grown via e-beam evaporation.

	Thickness (nm)	Roughness (nm)	Density (g/cm^3)
103.9 nm SiO_2	104.5311	0.3626	1.69738
204.7 nm SiO_2	203.8994	0.741	2.04566

Table 4.4: Thickness, roughness and densities calculated from XRR curve fitting

materials.

The results of the curve fitting performed using the Advanced Material Analysis and Simulation Software for the XRR spectra of selected SiO_2 thin film thicknesses as shown in Figure 4.9 and Figure 4.10 are presented in Table 4.4. The thicknesses also match the previously calculated values in the direct and Fourier method.

The calculated roughness for both films is also less than 1 nm, signifying the excellent quality and sharp interface of the deposited film required for multilayer antireflective coatings. Again, the densities of the films are generally lower than the expected density for SiO_2 as used for the QCM input ($\rho = 2.65$), and there is a much larger variation in densities between the two films compared to that for a-Si. This difference is also likely caused by the increased porosity of the films during deposition resulting from the absence of substrate heating, compounded by the previously mentioned difficulty in maintaining the stable evaporation of SiO_2.

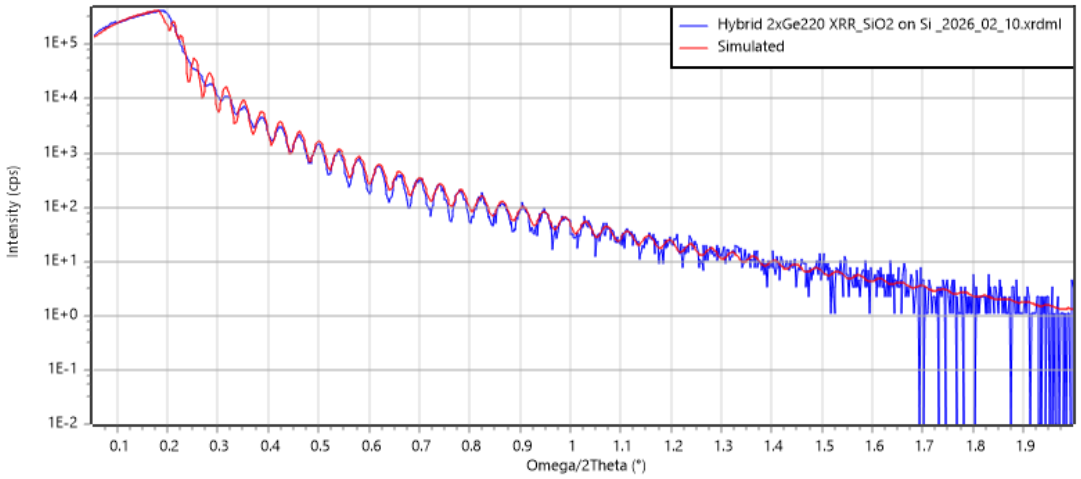


Figure 4.9: XRR curve fit to 103.9 nm SiO_2 thin film

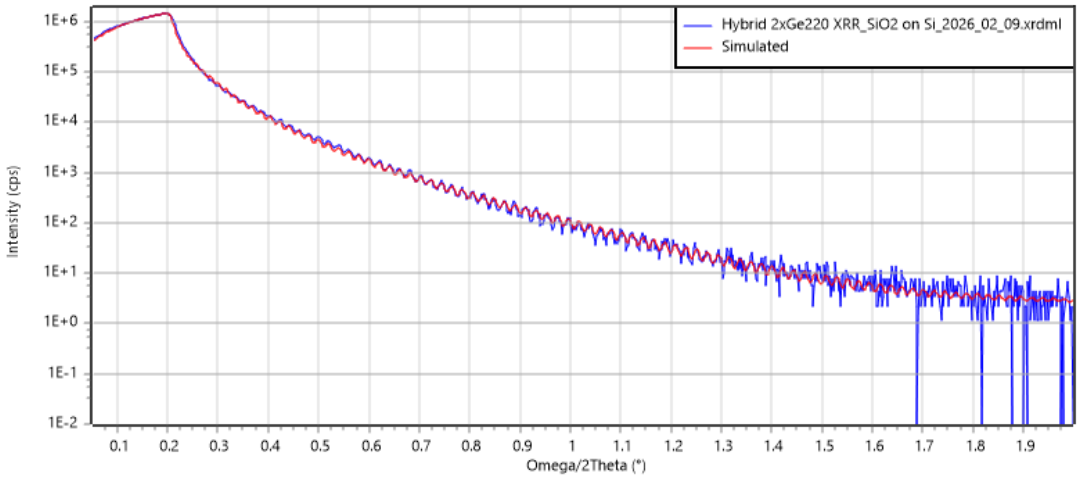


Figure 4.10: XRR curve fit to 204.7 nm SiO_2 thin film

4.2 Refractive index of a-Si thin films

Figure 4.11 shows the UV-Vis transmittance spectra of a 600 nm a-Si thin film deposited on a glass substrate together with the upper and lower envelopes as calculated using a modified version of the envelope method described in section 2.4. The equations of the exponential functions fitted to the extrema points listed in Table 4.5 using `scipy.optimize.curve_fit` are

$$y = 0.7848 - 250.7271e^{-x/103.5957} \tag{4.3}$$

$$y = 0.4543 - 464.5691e^{-x/83.2507} \tag{4.4}$$

for the upper and lower envelopes respectively. The last maxima at $\lambda = 655$ nm is excluded from further calculations due to the mismatch in expected value for m. The envelope method in section 2.4 specifies that succeeding extrema must correspond to half-integer steps, but the calculated value at this wavelength is $m = 5.0$ instead of a value close to the half-integer 5.5. This likely means that the envelope functions are no longer accurate at this wavelength. The minima at $\lambda = 613$ nm is also excluded because the calculated N is negative, caused by the crossing of the upper and lower envelopes. More fringes in the transmission spectra would probably result in more accurate envelopes which satisfy the necessary conditions outlined in section 2.4.

The equation for the refractive index of the a-Si thin film according to the Cauchy Equation 2.31 fitted to the remaining four data points of the refractive indices obtained from the envelope method using `scipy.optimize.curve_fit` is

$$n(\lambda) = (2.8716 \pm 0.0347) + \frac{38061.0 \pm 22671.1}{\lambda^2} \tag{4.5}$$

with $R^2 \approx 0.5849$ and the graph of the resulting refractive index versus wavelength is shown in Figure 4.13. The low R^2 is caused by the refractive index at 871 nm deviating from the trend of increasing refractive index for decreasing wavelength, likely due to the transmission envelope.

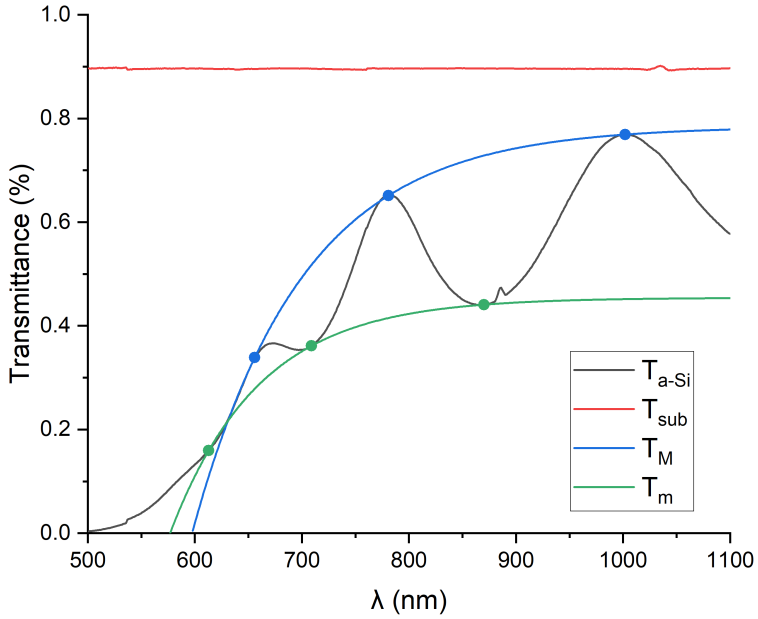


Figure 4.11: The transmission curves for the 600 nm a-Si thin film T_{a-Si} and the glass substrate T_{sub} together with the upper T_M and lower envelopes T_m for the envelope method. The envelopes cross at $\lambda = 630$ nm, likely caused by the very few data points used for fitting as highlighted.

λ (nm)	T_{a-Si}	T_{sub}	T_M	T_m	n_s	N	n_i	m	n_f
maxima									
1002	0.769	0.896	0.769	0.452	1.62	4.776	3.04	3.6	2.92
782	0.653	0.896	0.653	0.416	1.61	4.614	2.99	4.6	2.93
655	0.335	0.335	0.335	0.276	1.61	3.835	2.70	5.0	
minima									
871	0.441	0.896	0.729	0.441	1.61	4.682	3.01	4.2	2.90
709	0.361	0.896	0.518	0.361	1.61	4.491	2.95	5.0	2.95
613	0.160	0.896	0.123	0.167	1.61	-5.162			

Table 4.5: Relevant values for the modified envelope method of the a-Si thin film, with $d = 600$ nm. The calculations are not continued for $\lambda = 655$ nm since $m = 5.0$ instead of 5.5 when considering fringes, and for $\lambda = 613$ nm since $N < 0$.

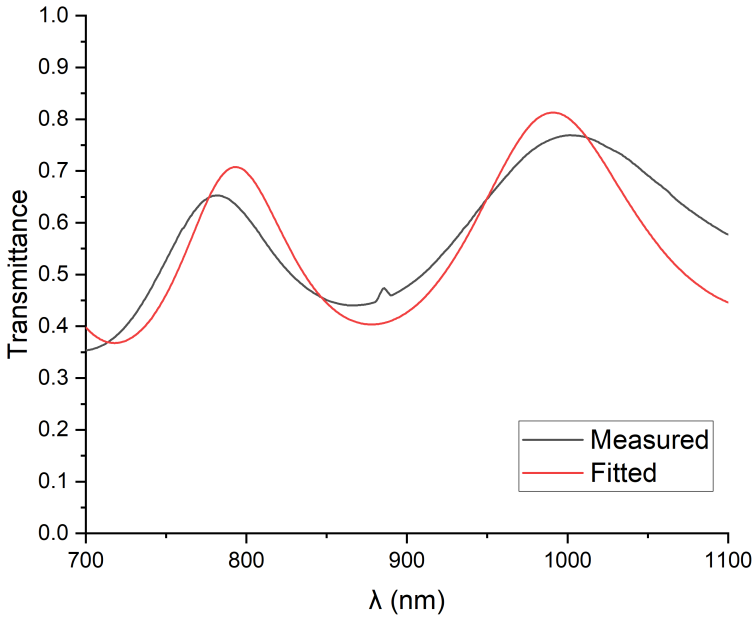


Figure 4.12: Direct fitting of transmittance from the transfer matrix method for the a-Si thin film. Values lower than 700 nm are excluded in order for the model to converge. The refractive index equation corresponding to the fitted curve is Equation 4.6

On the other hand, direct curve fitting of the Cauchy equation to the transmission spectra using the transfer matrix method and `scipy.optimize.curve_fit` as shown in Figure 4.12 gives the following equation for refractive index

$$n(\lambda) = (3.2971 \pm 0.0128) + \frac{0 \pm 9886.9}{\lambda^2} \tag{4.6}$$

as graphed in Figure 4.13. Values lower than 700 nm had to be excluded from the fitting in order for the model to converge properly, which means that the refractive index values corresponding to wavlengths lower than 700 nm are extrapolated.

The refractive indices of the a-Si thin film calculated for both methods are quite consistent, differing by around 0.25, and comparable to the literature values obtained by Swanepoel for a-Si at longer wavelengths[33]. Furthermore, the calculated a-Si thin film refractive index is lower than that of crystalline silicon, which has a refractive index of 3.5 to 4.25 across the range of wavelengths as measured by Schinke [50]. This is likely due to the lower density and less ordered structure of the a-Si thin film compared

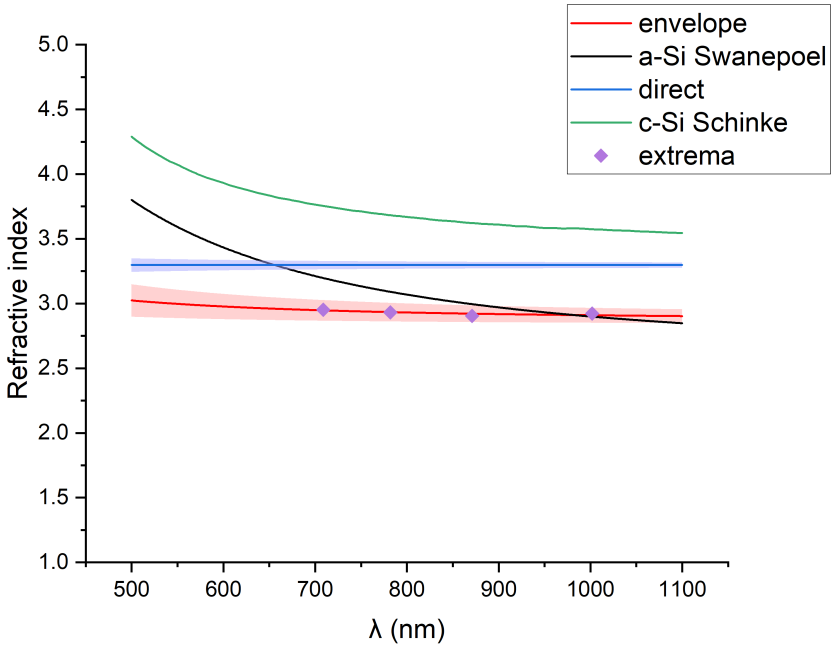


Figure 4.13: Refractive index of the a-Si thin film from 500 nm to 1100 nm as calculated from the envelope and direct fitting methods compared to values obtained by Swanepoel[33] and Schinke[50].

to crystalline silicon, and supports the previous findings in section 4.1. However, the refractive index appears somewhat flat over the wavelength range instead of increasing at shorter wavelengths. This difference from the expected behavior at shorter wavelengths is probably caused by the loss of information due to the fitting procedures used.

In the modified envelope method, the number of data points used to fit the transmission envelopes is too few, which results in less accurate transmission envelopes and consequently refractive index values. On the other hand, for the direct fitting method using the transfer matrix method, the accuracy of the calculated value depends on the quality of the fit to the experimental data and is also limited to the 700 nm to 1100 nm wavelength range, consequently extrapolation to shorter wavelengths may no longer be accurate.

4.3 Refractive index of SiO_2 thin films

Figure 4.14 shows the reflectance spectra of a 990 nm SiO_2 thin film deposited on a Si substrate after deconvolution using a mirror. For the extrema method there are six points

λ_0 (nm)	R	λ_1 (nm)	n_1	λ_2 (nm)	n_2
minima					
530.0	0.052	587.5	1.50	676.3	1.50
645.0	0.040	733.8	1.50		
822.5	0.043				
maxima					
577.5	0.154	648.8	1.47	765.0	1.48
720.0	0.140	836.3	1.48		
952.5	0.216				

Table 4.6: Refractive indices and wavelengths for the SiO_2 thin film at specified extrema points

that can be used for the calculation of the refractive index of the film using Equation 2.8; they are listed in Table 4.6 together with the corresponding refractive indices and the new average wavelengths. Fitting the resulting data points to the Cauchy Equation 2.31 using `scipy.optimize.curve_fit` as shown in the green points and blue curve in Figure 4.15 gives the following equation:

$$n(\lambda) = (1.431606 \pm 0.012136) + \frac{9540.05 \pm 4793.93}{\lambda^2}. \tag{4.7}$$

On the other hand, direct fitting using `scipy.optimize.curve_fit` and the transfer matrix method to fit the Cauchy equation directly to the reflectance data results in the following equation for the refractive index:

$$n(\lambda) = (1.431606 \pm 0.012136) + \frac{9540.05 \pm 4793.93}{\lambda^2} \tag{4.8}$$

as shown in the red curve in Figure 4.15

The refractive indices of the SiO_2 thin film obtained from both methods are comparable to the literature values obtained from Gao[51]. The extrema method has a slightly higher value versus the literature and a much larger error range, whereas the direct method has a slightly lower value and smaller error range. The increased uncertainty in refractive index values for the extrema method compared to the direct method is likely caused by the few data points used for fitting in the extrema method compared to the

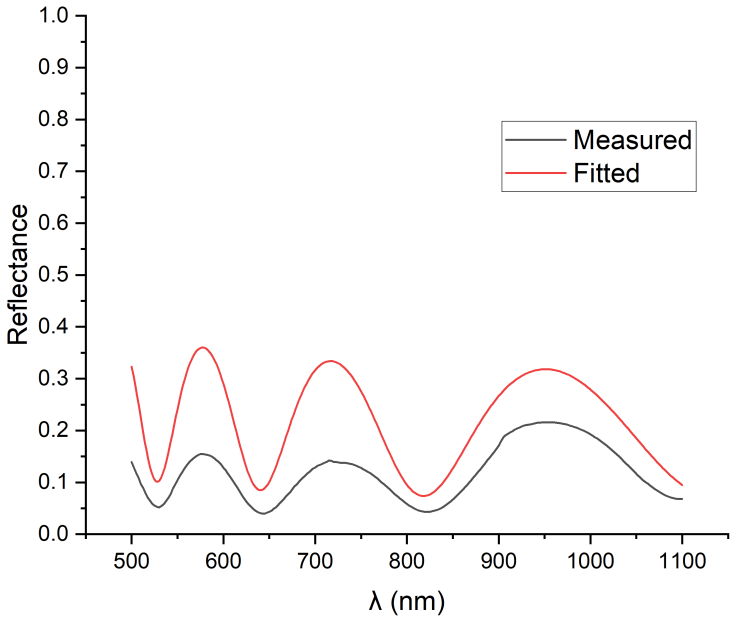


Figure 4.14: Deconvoluted reflectance spectra of 990 nm SiO_2 thin film together with the resulting reflectance from direct fitting of the Cauchy equation using the transfer matrix method.

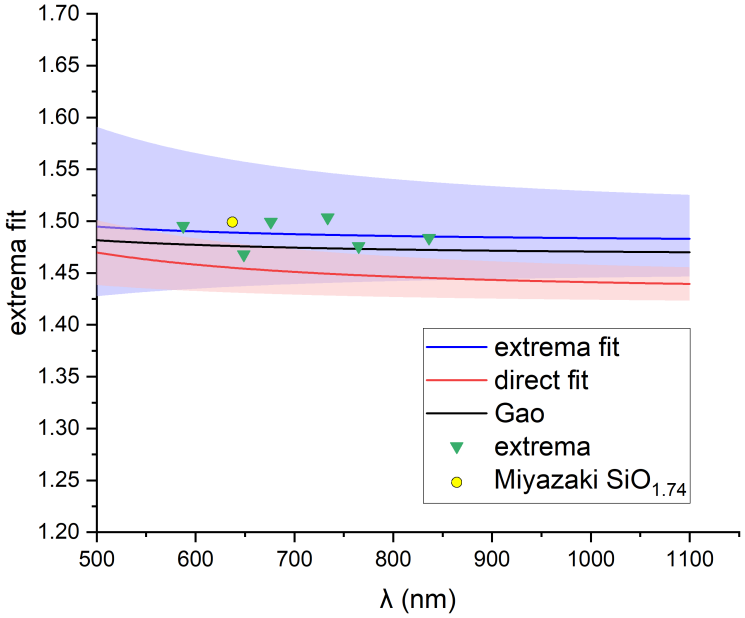


Figure 4.15: Refractive index of the SiO_2 film for the extrema and direct methods compared to the values obtained by Gao for SiO_2 and Miyazaki for $SiO_{1.74}$. The high uncertainty for the extrema method may be caused by the few data points fitted.

direct method.

The slightly lower refractive index value calculated through the direct method may be caused by multiple factors, such as increased porosity of the film, or poor fitting through the transfer matrix method. The films appear to be close to stoichiometric SiO_2 even though there was no reactive gas used during the deposition process. Miyazaki et al. deposited $SiO_{1.74}$ thin films and found that the refractive index at 633 nm was 1.50 [52], which is higher than the measured refractive index for both methods. This increased porosity may also account for the lower density of the SiO_2 thin film as measured by XRR previously compared to bulk SiO_2.

4.4 Reflectance of a-Si/SiO_2 antireflective coatings

Figure 4.16 shows the reflectance spectra after deconvolution relative to a mirror of a-Si/SiO_2 antireflective coatings with layer thicknesses of 53 nm /14 nm /206 nm, 48 nm /19 nm /208 nm, and 49 nm /24 nm /251 nm for the SiO_2/a-Si/SiO_2 layers together with the simulated reflectance from the TMM using the refractive indices obtained in the previous sections. The simulated reflectance spectra for the first antireflective coating has a value less than 1 % around 656 nm, the simulation of the second antireflective coating has a minimum value less than 1 % at 684 nm, and the third antireflective coating has less than 1 % at 778 nm. The measured reflectance spectra for the antireflective coatings shows the minimum values less than 4 % for the first antireflective coating at 662 nm, 2.3 % for the second antireflective coating at 716 nm, and less than 1 % at 835 nm for the third ARC.The values of the reflectance at 780 nm are around 24 % for the first ARC, 10 % for the second ARC, and 7 % for the third ARC. Table 4.7 summarizes the relevant information of the antireflective coatings being studied. The blank Si wafer exhibits a reflectance ranging from 50 % to 75 % across the wavelength range of 450 nm to 1000 nm.

The multilayer coatings deposited using electron beam deposition have reduced values of reflectance compared to the bare wafer, proving their effectiveness as antireflective coatings. The mismatch in the minimum reflectance values obtained experimentally

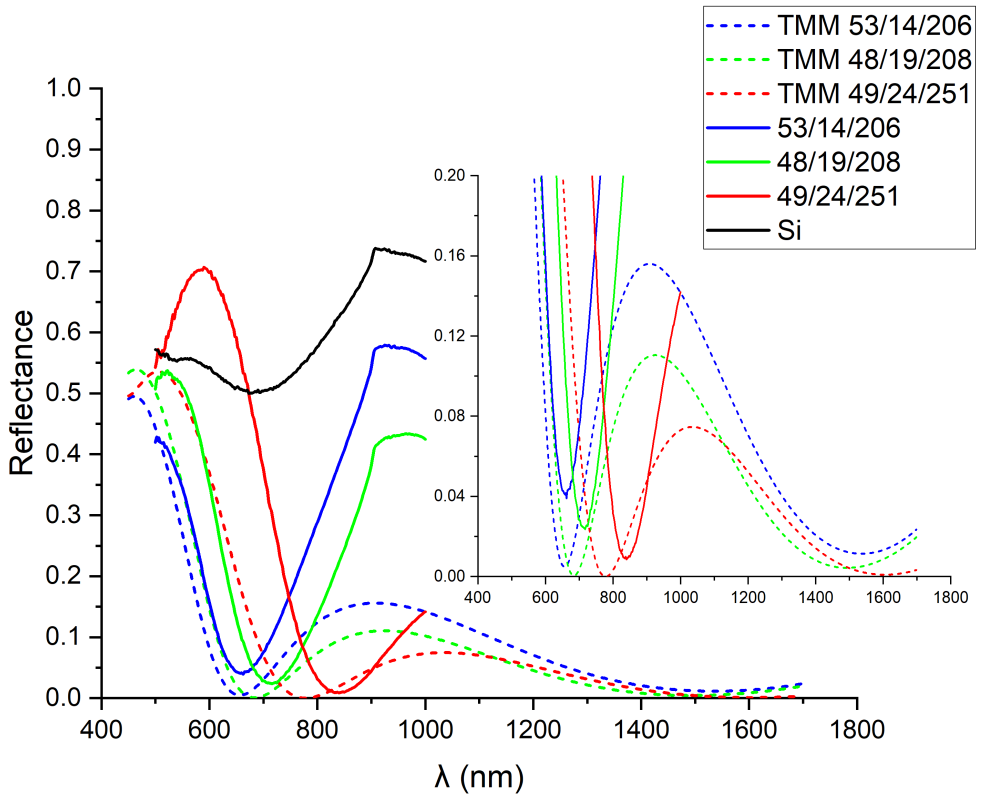


Figure 4.16: Reflectivities of antireflective coatings with varying layer thicknesses simulated using the transfer matrix method and measured experimentally. The reflectance of a bare Si wafer is included for comparison.

	1st minimum		2nd minimum		R at 780 nm
	λ (nm)	R	λ_0 (nm)	R	
53 nm/14 nm/206 nm TMM	656	0.00492	1533	0.0114	0.10945
53 nm/14 nm/206 nm	663	0.03906			0.24194
48 nm/19 nm/208 nm TMM	684	0.00049	1482	0.0043	0.05995
48 nm/19 nm/208 nm	718	0.02367			0.09677
49 nm/24 nm/251 nm TMM	778	0.00003	1606	0.0009	0.00006
49 nm/24 nm/251 nm	840	0.00858			0.06700
Si					0.55323

Table 4.7: Simulated and experimental reflectance of the antireflective coatings and Si wafer at different wavelengths

versus the simulation is due to the nature of the experimental setup, which measures the reflectance of the sample relative to the mirror. More accurate reflectance values would be obtained from absolute measurements which consider the total amount of light incident and reflected from the sample. Interestingly, the deviation of wavelengths between the measured minima of the films and that of their simulated counter parts increases as the target wavelength increases; a difference of 7 nm at a target wavelength of 656 nm, which increases to 34 nm at a target wavelength of 684 nm, and finally 62 nm at a target wavelength of 656 nm. This shift in values is likely caused by the inaccuracy in measuring the refractive index of the thin films which were previously deposited, especially for a-Si. The refractive index values calculated may have been accurate at the lower wavelengths but increasingly erroneous for longer wavelengths, which corroborates the observation in section 4.2 where the index does not monotonically decrease with increasing wavelength.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

In this study, an antireflective coating made of multiple layers of a-Si/SiO_2 with a minimum close to the wavelengths at which terahertz photoconductive antennas are optically pumped has been presented. The transfer matrix model was used to obtain design parameters which supplied information for the thicknesses to be used in the fabrication with electron beam physical vapor deposition. The refractive index data used for the transfer matrix model was obtained from measurements on single films.

The thicknesses for a-Si and SiO_2 obtained using x-ray reflectivity showed that the quartz crystal microbalance readings set at the reference material parameters generally underestimate the actual thicknesses of the deposited films, but follows a linear relationship with them. The densities of the evaporated films are generally lower than that of the bulk. The refractive index of the a-Si film measured via the envelope method compared to the literature shows a deviation from the usual monotonically decreasing value with increasing wavelength. For the refractive index of the SiO_2 film obtained via reflection spectroscopy, the values are higher than the reference, which may be explained by a change in stoichiometric ratio of the film. The measured reflectance of the multilayer coatings generally match the expected minima from the transfer matrix model simulations, although the inaccuracy increases at longer wavelengths.

Recommendations for further studies into the topic include more accurate measurements of the material refractive indices in order to obtain better estimations from the transfer matrix method. Improvements to the quality of the films can also be studied, particularly if substrate heating during growth causes the index to match literature values better. A broader measurement range that includes the infrared region at 1550 nm would also yield more information about the properties of the antireflective coating.

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Appendices

.1 Transfer Matrix Method python code

```
# TMM for 9-layers - MJ Hurben Dec 2024
# Modified for 1550 nm and 780 nm optimized SiO2/Si ARC
    - GCC 2026
import numpy as np
import matplotlib.pyplot as plt
d2r = np.pi / 180  # Degrees to radians factor
na=1               # Index of air
n_layers=3         # Number of layers

# Define matrices of refractive index n and thickness d,
    modify functions and arrays as needed

# try to use complex refractive index n(wl)+k(wl)j for
    formula?

def ns(wl): # Define ns as a function of wl based on
    index of substrate material Si from 430 nm
      return 4.0333E-12*wl**4 - 1.7984E-08*wl**3 +
          2.9993E-05*wl**2 - 2.2437E-02*wl + 9.9702

#     return -8.974E-15*wl**5 + 5.299E-11*wl**4 -
    1.228E-07*wl**3 + 1.398E-04*wl**2 - 7.857E-02*wl +
    2.103E+01 # In52Ga48As

def n1(wl): # Define n1 as a function of wl based on
    refractive index of 1st layer material SiO2
```

```
    return np.sqrt(20201856809.3-22607182010.85*wl**2/
        (wl**2-0.1063836985797) +2405325203.746*wl**2/
        (wl**2-0.9998803009815))

def n2(wl):  # Define n2 as a function of wl a-Si from
    430 nm
    return np.sqrt(14.84529+(7067090000000)*wl**2/
        (wl**2-(-21244900000000000)) +(-10.41841)*wl**2/
        (wl**2-(-190469.027)))

def refractive_index_array(wl): # Add corresponding
    variables and links to functions for each layer
    n_s = ns(wl)
    n1_val = n1(wl)
    n2_val = n2(wl)
    return np.array([na, na, na, na, na, na, n1_val,
        n2_val, n1_val, n_s])   # If < 9 layers, use na as
        index of 'missing' layers


d=np.array([1,1,1,1,1,1,206,14,53,1000]) # d-Matrix
    values are in nm; initial air layers and final glass
    layer arbitrary

def PS(nva,dva,wl): # Phase Shift calculation

    def M(nA,nB): # Transfer matrix for interface
        r=(nA-nB)/(nA+nB)
        t=1+r
```

```
        d=1/t
        offd=r/t
        m=np.array([[d,offd],[offd,d]])
        return m

    def P(L,n):  # Transfer matrix for propagation
        L=L*n*2*np.pi
        v=complex(np.cos(L),-np.sin(L))
        p=np.array([[v,0],[0,v.conjugate()]])
        return p

    W=M(nva[8],nva[9]) # Interface of last layer and
        glass
    for i in range(8,-1,-1):
        W=np.matmul(P(dva[i]/wl,nva[i]),W)
        if i>0:
            W=np.matmul(M(nva[i-1],nva[i]),W)
        else:
            W=np.matmul(M(na,nva[0]),W)  # Interface of
                glass and first layer
        r=W[1,0]/W[0,0]
    return r


d_sweep=np.array([1,1,1,1,1,1,206,14,53,1000])    #
    Matrix for thickness sweep
d_optimal=np.array([1,1,1,1,1,1,206,14,53,1000])

for layer in range(int(9-n_layers),9):    # Rough sweep
```

```
    (100 nm range 5 nm step) for best thickness
      original_val=d_sweep[layer]
      base_reflectance= abs( PS(
         refractive_index_array(1550), d_sweep,1550))**2 +
         abs( PS( refractive_index_array(780)
         ,d_sweep,780)) **2
      for sweep_size in range(-50,55,5):
          d_sweep[layer]=original_val+sweep_size
          r_init_1550=
             PS(refractive_index_array(1550),d_sweep,1550)
          r_init_780=
             PS(refractive_index_array(780),d_sweep,780)
          if (abs(r_init_1550)**2+abs(r_init_780)**2)
             <base_reflectance:   # try reducing sum of
             base_reflectance at 1550 and 780?
               d_optimal[layer]=d_sweep[layer]
               base_reflectance =
                  (abs(r_init_1550)**2+abs(r_init_780)**2)
      d_sweep[layer]=d_optimal[layer]

for layer in range(int(9-n_layers),9):   # Fine sweep
    (10 nm range 1 nm step) for best thickness
      original_val=d_sweep[layer]
      base_reflectance = abs( PS(
         refractive_index_array(1550), d_sweep,1550)) **2+
         abs( PS( refractive_index_array(780),
         d_sweep,780)) **2
      for sweep_size in range(-5,6):
          d_sweep[layer]=original_val+sweep_size
```

```
            r_init_1550 =
               PS(refractive_index_array(1550),d_sweep,1550)
            r_init_780 =
               PS(refractive_index_array(780),d_sweep,780)
            if (abs(r_init_1550)**2+abs(r_init_780)**2)
               <base_reflectance:    # try reducing sum of
               base_reflectance at 1550 and 780?
                d_optimal[layer]=d_sweep[layer]
                base_reflectance =
                   (abs(r_init_1550)**2+abs(r_init_780)**2)
        d_sweep[layer]=d_optimal[layer]

wls=[] # Array of wavelengths
T=[] # Array of T values
R=[] # Array of R values
R_opt=[] # Array of optimized R values

for wl in range(430,1700,2):
        n=refractive_index_array(wl)
        r=PS(n,d,wl)                    # reflectance of initial
           values
        r_opt=PS(n,d_optimal,wl)    # reflectance of
           optimized values
        wls.append(wl)
        R.append(abs(r)**2)
        R_opt.append(abs(r_opt)**2)
#       T.append(1-abs(r)**2)

print(d_optimal)
```

```
export_array = np.column_stack((wls, R, R_opt))
np.savetxt("tmm.csv", export_array, delimiter=",")

plt.grid(True)
plt.plot(wls, R, label="Initial")
plt.plot(wls, R_opt, label="optimized")
plt.xlim(430, 1700)
plt.ylim(0.00,1)
plt.xlabel('Wavelength (nm)')
plt.ylabel('Reflectance R')
plt.legend(loc="upper right")
```

.2 Fitting code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit


def tmm_3layer_reflectance(wavelengths, n1_fn, n2_fn,
    n3_fn, t1, t2, t3, n0=1.0, n_sub=1.5):
    """Calculates reflectance spectrum R(lambda) for a
       3-layer thin film stack."""
    wl = np.asarray(wavelengths, dtype=float)

    n0_arr = n0(wl) if callable(n0) else
       np.full_like(wl, n0, dtype=complex)
    nsub_arr = n_sub(wl) if callable(n_sub) else
```

```
    np.full_like(wl, n_sub, dtype=complex)

n_funcs = [n1_fn, n2_fn, n3_fn]
thicknesses = [t1, t2, t3]

M = np.zeros((len(wl), 2, 2), dtype=complex)
M[:, 0, 0] = 1.0
M[:, 1, 1] = 1.0

for n_fn, t in zip(n_funcs, thicknesses):
    n_layer = n_fn(wl) if callable(n_fn) else
        np.full_like(wl, n_fn, dtype=complex)
    delta = 2.0 * np.pi * n_layer * t / wl

    cos_d = np.cos(delta)
    sin_d = np.sin(delta)

    m_layer = np.zeros((len(wl), 2, 2),
        dtype=complex)
    m_layer[:, 0, 0] = cos_d
    m_layer[:, 0, 1] = -1j / n_layer * sin_d
    m_layer[:, 1, 0] = -1j * n_layer * sin_d
    m_layer[:, 1, 1] = cos_d

    M = M @ m_layer

B = M[:, 0, 0] + M[:, 0, 1] * nsub_arr
C = M[:, 1, 0] + M[:, 1, 1] * nsub_arr
```

```
    r = (n0_arr * B - C) / (n0_arr * B + C)
    return np.abs(r)**2


def fit_n2_parameters(csv_filepath, n1_fn, t1, t2, t3,
    n0=1.0, n_sub=1.5, p0=[2.0, 5000.0]):
    """
    Fits parameters a and b for n2(wl) = a + b/wl**2
    from CSV data.
    Assumes layer 3 refractive index equals layer 1 (n3
    = n1).

    Parameters:
    - csv_filepath : path to CSV file containing columns
    'wavelength' and 'reflectance'
    - n1_fn        : fixed refractive index function or
    constant for layer 1 (and layer 3)
    - t1, t2, t3   : layer thicknesses (nm)
    - n0           : ambient refractive index
    - n_sub        : substrate refractive index
    - p0           : initial guess [a_init, b_init] for
    layer 2
    """
    # 1. Load experimental CSV data
    df = pd.read_csv(csv_filepath)
    wl_data = df["wavelength"].values
    R_data = df["reflectance"].values

    # Layer 3 has the same refractive index as Layer 1
```

```
n3_fn = n1_fn

# 2. Define wrapper model for curve_fit (n2 is the
    unknown function)
def model_reflectance(wl, a, b):
    n2_fn = lambda w: a + b / (w**2)
    return tmm_3layer_reflectance(wl, n1_fn, n2_fn,
        n3_fn, t1, t2, t3, n0=n0, n_sub=n_sub)

# 3. Perform optimization with physical bounds (a >
    1, b >= 0)
bounds = ([1.0, 0.0], [4.0, 1e6])
popt, pcov = curve_fit(model_reflectance, wl_data,
    R_data, p0=p0, bounds=bounds)

a_fit, b_fit = popt
a_err, b_err = np.sqrt(np.diag(pcov))

print("---■Fit■Results■for■Layer■2■---")
print(f"a■=■{a_fit:.5f}■+/-■{a_err:.5f}")
print(f"b■=■{b_fit:.2f}■+/-■{b_err:.2f}■nm^2")

# 4. Plot fitted curve vs experimental data
R_fit = model_reflectance(wl_data, a_fit, b_fit)

plt.figure(figsize=(8, 5))
plt.plot(wl_data, R_data, "o", label="CSV■Data",
    color="crimson", alpha=0.6, markersize=4)
```

```
    # Raw string with .format() avoids LaTeX backslash
        errors in f-strings
    label_text = r"Fit (${n_2} = {:.3f} + \frac{{{:.1f}
        }}{{\lambda^2}}$)".format(a_fit, b_fit)
    plt.plot(wl_data, R_fit, "-", label=label_text,
        color="black", lw=2)

    plt.xlabel("Wavelength (nm)")
    plt.ylabel("Reflectance")
    plt.title("TMM Parameter Fitting for Layer 2 (${n_3} =
        n_1$)")
    plt.grid(True, linestyle="--", alpha=0.5)
    plt.legend()
    plt.tight_layout()
    plt.show()

    return a_fit, b_fit


# --- Usage Example ---
if __name__ == "__main__":
    sample_wl = np.linspace(400, 800, 50)

    # Known layer 1 function (n3 will automatically copy
        this)
    n1_known = lambda w:
        np.sqrt(20201856809.3-22607182010.85*w**2/
        (w**2-0.1063836985797) +2405325203.746*w**2/
        (w**2-0.9998803009815))
```

```
    # Target values for layer 2: a = 2.1, b = 12000
    n2_target = lambda w: 2.60 + 300000.0 / (w**2)

    # Known substrate function
    n_si = lambda w: 4.0333E-12*w**4 - 1.7984E-08*w**3 +
        2.9993E-05*w**2 - 2.2437E-02*w + 9.9702

    # Generate synthetic experimental data (n1 = n3 =
        n1_known)
    R_synthetic = tmm_3layer_reflectance(
        sample_wl, n1_known, n2_target, n1_known,
            t1=208, t2=19, t3=48, n0=1.0, n_sub=n_si
    )
    R_synthetic += np.random.normal(0, 0.003,
        size=len(sample_wl))  # Add noise

    # Save simulated CSV
#     pd.DataFrame({"wavelength": sample_wl,
    "reflectance":
    R_synthetic}).to_csv("measured_data.csv", index=False)

    # Run parameter fitting for layer 2
    a_opt, b_opt = fit_n2_parameters(
        csv_filepath="measured_data2.csv",
        n1_fn=n1_known,
        t1=208,
        t2=19.0,
        t3=48,
```

```
        n0=1.0,
        n_sub=n_si,
        p0=[2.6, 300000.0]  # Initial guess
    )
```